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An explicit exploration strategy for evolutionary algorithms

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ABSTRACT

Bio-inspired algorithms are a trending topic in the field of optimization. In general, evolutionary algorithms start their search process by generating a random population. Although there are some works in the literature that propose a mechanism to generate the initial population of a particular algorithm, there is no general approach to improve exploration in evolutionary algorithms. In this work, a versatile strategy is proposed for improving the exploration of any evolutionary algorithm. An exhaustive set of experiments was designed by applying 4 state-of-the-art algorithms to a well-known benchmark of continuous functions. A comparison based on statistical tests was made to demonstrate the performance of the proposed policy. The results indicate that the explicit exploration strategy is effective.

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1. Introduction

The goal of evolutionary algorithms (EAs) is to find high quality solutions for optimization problems at a reasonable computational cost. Thus, the researchers from the evolutionary computation community have proposed several methodologies in order to make an effective and efficient search. Although such proposals are based on different paradigms, in most cases their performance highly depends on their capacity to adequately manage their own search procedure. In this work, a new and general search policy is introduced for helping manage the procedure of searching in EAs. The proposed search policy does not modify the main paradigm behind of each EA and can be considered as a complementary resource.

The existing paradigms in EAs have been able to provide reasonable solutions to complex problems by using their corresponding specific search procedure. Although this diversity of paradigms has created a great number of different search procedures, the working of an evolutionary algorithm can be broadly described by the important role of two search aspects: the exploration and the exploitation [1]. In general, the exploration component of the search process allows to discover promising regions and the exploitation component of the search process allows to obtain high quality solutions. Thus, the search process done by EAs can be understood by means of the exploration and the exploitation components. Moreover, as stated in [2], the search process in EAs is often referred to in terms of a trade-off between exploration and exploitation.

The main goals of the exploration and the exploitation can be identified and deduced from the related literature, such as the works of [2–9], and [10]. These goals are next described. The exploration has the purpose of visiting, as many as possible, new and different regions from the search space in order to discover and provide good quality solutions. On the other hand, the purpose of the exploitation is to visit the promising regions already identified around the good quality solutions in order to further refining them and to obtain better solutions. Therefore, the exploration helps escape from local optima and the exploitation helps improve the quality of solutions.

It is worth mentioning that the beneficial aspects of the exploration and the exploitation are always desired but it can become difficult to guarantee them. Thus, the lack of equilibrium between the exploration and the exploitation can lead to undesirable consequences such as a premature convergence or a high computational cost. This concern has motivated researchers to analyze the working mechanisms of evolutionary algorithms with the aim to achieve an optimal equilibrium between the exploration and the exploitation.

1.1. Related work

The general performance of EAs depends on their ability to find promising sets of individuals and their capacity to improve the quality of the selected solutions. In the following, we present different contributions related to the proposals for achieving an optimal balance of exploration and exploitation in EAs.

The desired equilibrium between exploration and exploitation, as stated in [11], can be achieved by controlling the population diversity and this idea was the basis for proposing a parameter adaptation for Differential Evolution algorithms. For [12],

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the need to quantify the search behavior of a Particle Swarm Optimization algorithm is the starting point for determining the usefulness of several diversity measures in quantifying the exploration/exploitation of a particle swarm algorithm. In [13], it is recognized that the balance between exploration and exploitation depends on the characteristics of different problems and an auto-tuning strategy by using fuzzy logic control is proposed for Genetic Algorithms (GAs). In [14], an entropy-driven approach is introduced to control balance between exploration and exploitation in Genetic Algorithms. For exploration and exploitation analysis, as well as the possibility of using this analysis for evolutionary algorithm designing, an ancestry tree-based approach is introduced in [15].

The exploration–exploitation ratio is taken into account in [9] for designing an algorithm based on the evolution process of the three states of matter: the gas state, the liquid state and the solid state. This proposal relates the behavior of the states of matter with three different search levels: intensified exploration (gas state), transition from exploration to exploitation (liquid state) and pure exploitation (solid state). For dynamic optimization problems, in [16], two equally sized populations are used for making explicit the two components that guide any search technique: the first population performs exploration while the second population is responsible for exploitation. A nonlinear control strategy over the distance parameter is proposed in [17] for improving the exploration and exploitation capabilities of the Whale Optimization Algorithm. In [18], a novel survival analysis method is proposed to guide the maintenance of the exploration and exploitation trade-off in multiobjective evolutionary algorithms. In [19], a new selection operator for GAs is proposed for giving a better trade-off between exploration and exploitation.

According to the above, there is evidence to show that EAs have incorporated different strategies for controlling and improving their balance between exploration and exploitation. Furthermore, this evidence shows that each evolutionary algorithm has been provided with particular strategies for tuning its own control parameters and evolutive operators in order to successfully solve specific optimization problems.

The previous approaches control the population diversity in order to achieve an adequate equilibrium between exploration and exploitation. These approaches require mechanisms for detecting when to increase the population diversity and how to do it.

1.2. The initial population

The experimental results obtained by [20] provide evidence for concluding that the main cause of the imbalance between exploration and exploitation can be: (1) the combination of operators and settings of the EA, or (2) the fitness landscape. Thus, besides the specific working mechanisms of each EA, the characteristics of the optimization problem play an important role for obtaining an adequate equilibrium between exploration and exploitation.

In general, the initial population of an evolutionary algorithm is generated for providing the first information about the fitness landscape. As a consequence, the set of the initial individuals is an important source of information for the exploration stage and it may be influential in the following steps of the search process. For instance, [21,22] are two contributions related to the study of initialization techniques and their relationship to the performance and behavior of search processes. In [21], the association between the uniformity of the population set and the dimension of the search space is studied. In [22], a kernel density is proposed for generating an initial population with desirable properties of uniformity and randomness. The proposal presented in [22] compares several initialization techniques and shows how the behavior of an evolutionary algorithm (CMA-ES) can be influenced by the technique used for generating the initial population. The interested reader in initialization techniques is referred to [23], where these techniques are presented in a comprehensive and systematic way. According to [23], from several existing initialization techniques, the stochastic technique known as Pseudo-Random Number Generator (PRNG) is widely used in EAs.

The initial population is the first source of information for many EAs about the fitness landscape. In general, it is expected a good quality from the information provided by the initial population. In order to avoid biased information from the optimization problem, the uniform probability distribution is commonly used as the initial searching distribution for the PRNG. Thus, the set of initial individuals comes from an uniform distribution without any preference or bias for specific regions in the search space. To the best of our knowledge, there has been paid little attention to the exploration process of generating the first population and its contribution to the effectiveness of EAs. In particular, as hypothesis, the required size of the initial population and the complexity of the fitness landscape could be concordant or related. This conjecture requires a deep study.

1.3. Contributions

This paper provides the following contributions.

1. A general strategy for reinforcing the initial exploration and to improve the search process. The proposed strategy is based on the extension of the initial exploration and can be implemented along with *any* evolutionary algorithm.
2. A new perspective about the population size and number of generations required by an adequate initial exploration. The proposed strategy discards a fixed size for the initial population and introduces the fitness distribution as an element of information about the quality of the exploration process.
3. A simple criterion for evaluating the fitness information provided by the initial exploration. This criterion is used for deciding when to stop the initial exploration.
4. A more informative set of parents about the fitness landscape is provided at the end of the proposed strategy. The quality of the selected individuals at the end of the exploration increases when the size of the initial population is great enough.

1.4. Structure of the manuscript

The rest of this paper is arranged as follows. In Section 2, the Explicit Exploration Strategy is explained. Section 3 provides details about the benchmark and the evolutionary algorithms applied to test the approach. Section 4 discusses the results obtained by each evolutionary algorithm applying the strategy. Finally, in Section 5 highlights about the Explicit Exploration Strategy are presented.

2. Explicit exploration strategy

The search process of EAs begins with an explicit exploration of the fitness landscape through the initial population. This explicit exploration samples the search space avoiding preference for specific regions by randomly generating individuals from the uniform probability distribution. Hence, it is desired that the explicit exploration provides enough information from the optimization problem in order to have a robust initial search. The more useful information from exploration, the more possibilities

of escaping from local optima and avoiding premature convergence. However, it must be considered that a search process only based on explicit exploration may reduce the efficiency of the algorithm. Due this, it is suggested in this work to stop the explicit exploration when the cumulative information about the fitness distribution is steady.

The goal of the Explicit Exploration Strategy (EES) is to obtain sufficient information about the optimization problem with the minimal number of individuals in the initial population. This can be achieved by taking into account the increase of information related to the added individuals in the initial population, across the first generations. Thus, an initial population will increase its size until there will not be a significant increment of information. It is worth to note that, for the EES, the initial population refers to all individuals randomly generated from the uniform probability distribution and the size of the initial population will increase along the first generations.

The fitness function is a deterministic rule that assigns a specific value to a given set of decision variables of an individual. However, in EAs, the initial exploration generates random values for the decision variables of the individuals. Thus, the fitness values of a random set of individuals are considered observations of a random variable. Hence, it can be stated that the fitness information provided by the initial population is random. Despite this randomness, the values of random quantities behave according to certain distribution. This means that a random set of individuals provides fitness information from a specific distribution. The estimation of this distribution and the size of the initial population are expected to be related. Thus, the size of the initial population will have an effect on the quality of the estimated distribution.

The following example illustrates the basics of the EES. An initial population is randomly generated with 200 individuals for optimizing the Ackley function in a twenty-dimensional search space. The description of the Ackley function can be consulted in Section 3.1. The Ackley function is used for evaluating the fitness of the 200 individuals from the initial population. The fitness information provided for this population is summarized by their 10th, 50th and 90th percentiles, also called 1st, 5th and 9th deciles. Next, the population size is increased by another 200 individuals randomly generated and its cumulative fitness information is summarized by their 10th, 50th and 90th percentiles. This process is repeated 48 times or, similarly, 48 generations until complete a cumulative population with 10,000 individuals. Thus, the 10th, 50th and 90th percentiles of the cumulative fitness information are registered in terms of the population size. Fig. 1(a) shows a descriptive relationship between the fitness information and the cumulative size of the initial population for the Ackley function.

The fitness information in Fig. 1(a) is shown by their 10th, 50th and 90th percentiles in terms of the initial population size or, equivalently, in terms of the number of generations. In Fig. 1, a number of 30 runs were done to provide the gray trajectories whereas the black trajectory corresponds to their average. The gray trajectories in Fig. 1(a) show, for any percentile, that there is more uncertainty with the fitness information provided by an initial population with a small size. Fig. 1(a) also shows the opposite relationship, an initial population with a large size provides fitness information with less uncertainty for any percentile. For this example, an approximate size of 5000 initial individuals provides similar information that initial populations with more than 5000 individuals. In terms of number generations, the first 24 generations of explicit exploration provides unsteady fitness information. Though the average of the information is practically the same for any size of the initial population, see black trajectories in Fig. 1(a), the variability of the fitness information has a decreasing behavior respect to the size of the initial population as shown by the gray trajectories.

The EES allows to select a more informative set of individuals once the explicit exploration has finished. In EAs, after the first generation, the successive generations require a set of individuals for generating the new population. This required set of parents comes from the previous generation and it is selected based on a criterion, except for the initial population where the set of parents is not required. In the above example, the 200 individuals with the best fitness are the selected population for each cumulative size of the initial population. For instance, when the size of the initial population is 200, the selected population equals the initial population. However, when the size of the initial population is greater than 200, the selected population is a proper subset of the initial population. Fig. 1(b) shows the 10th, 50th and 90th percentiles of fitness information related to the selected population in terms of the initial population size. Given that the goal is to minimize the Ackley function, it can be seen in Fig. 1(b) that the descending trajectories of fitness associated to the selected population indicate a quality increase for selecting the best individuals in the EES. This means that, at the end of several generations of the explicit exploration, the following search stage will be benefited with a more informative set of parents.

The EES is able to suggest a proper size of the initial population or, equivalently, the number of generations required for the explicit exploration. Fig. 1(a) shows that variability of the percentiles decreases as the size of the initial population increases. Hence, a comparison between the percentiles of two consecutive generations is useful for making a decision about to continuing or stopping the explicit exploration. It is suggested to add individuals in the initial population whether the compared percentiles of the fitness are not similar enough, otherwise, the recommendation is not to add individuals and take the last augmented population as the initial population.

An objective comparison between the percentiles of two consecutive generations can be done by establishing a dissimilarity criterion. The proposal in this work is the use of a quite simple criterion such as the relative Euclidean distance between quantiles of two consecutive generations. There are several sets of known quantiles such as quartiles, deciles and centiles. The number of the selected quantiles is related to the computational cost of the Euclidean distance. The quartiles provide the 25th, 50th and 75th percentiles, deciles provide the 10th, 20th, …, and 90th percentiles whereas centiles provide the 1st, 2nd, …and 99th percentiles. A comparison between deciles an centiles shows that centiles provide more knowledge of the distribution than deciles, but centiles involves more computational cost than deciles. A similar comparison can be made between quartiles and deciles. Thus, the set of deciles is chosen for calculating the relative Euclidean distance between generation $g-1$ and generation g, as shown in Eq. (1)

$$\frac{\sqrt{\sum_{i=1}^{9}(i\text{th decile}^{g-1} - i\text{th decile}^{g})^2}}{\sqrt{\sum_{i=1}^{9}(i\text{th decile}^{g-1})^2}}. \tag{1}$$

The relative Euclidean distance allows us to evaluate the increase of information when new individuals are added to the population. In case the fitness information were identical for two consecutive generations, then their corresponding deciles will be also identical and the numerator of Eq. (1) will be zero. However, when the generation g provides new information to the fitness distribution achieved in generation $g-1$, then the numerator of Eq. (1) will be greater than zero. It can be noted that the relative Euclidean distance measures the changes of fitness information in generation g respect to the fitness information in generation $g-1$. Though popular and easy to compute, the Euclidean distance belongs to a broad set of statistical equations for measuring dissimilarity between two probability distributions.

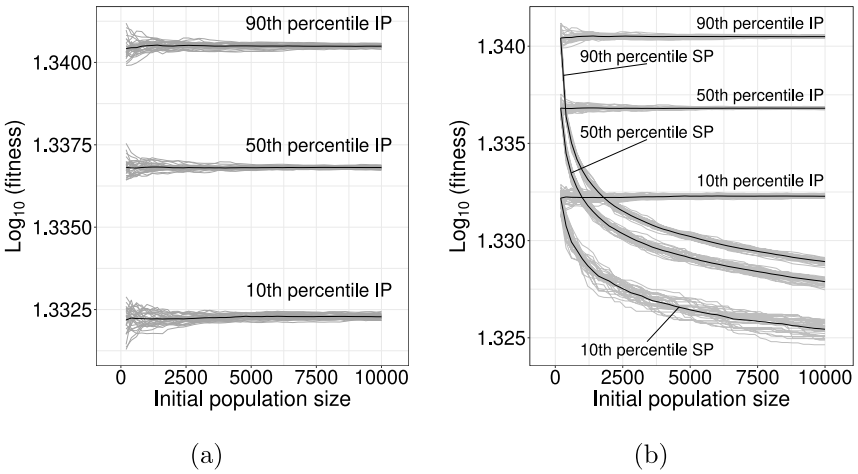


Fig. 1. Fitness trajectories for the Ackley function in the search space $[-100, 100]^{20}$ according to the size of the initial population. (a) 10th, 50th and 90th percentiles from the initial population (IP). (b) 10th, 50th and 90th percentiles from the selected population (SP).

Some statistical equations for measuring dissimilarities include the Kullback–Leibler divergence, Hellinger distance, Wasserstein distance, Kolmogorov–Smirnov statistic, Jensen–Shannon divergence, among others, and any of them can be used as a dissimilarity criterion for the EES. However, given that one of the purposes of this paper is to provide a simple approach for introducing the EES, the use and the comparison of sophisticated equations for measuring dissimilarities between fitness distributions go beyond the scope of this article. In summary, this work proposes the dissimilarity between two consecutive distributions as a way to measure the fitness information provided by new individuals. Further research can be done for analyzing the influence of different dissimilarities approaches.

For illustrative purposes, Fig. 2 shows the trajectories of the relative Euclidean distance applied to the previous example of the Ackley function. Fig. 2 shows the decreasing relationship between the relative Euclidean distance and the generations of explicit exploration, or equivalently, it shows the decreasing relationship of the relative Euclidean distance in terms of the size of the augmented initial population. The trajectories shown in Fig. 2 indicate a descending trend, which means that deciles of consecutive generations tend to be more similar among them. Besides, Fig. 2 also shows the inherent variability of the search process. Hence, if the relative Euclidean distance is less than a tolerance error, then the explicit exploration is stopped. However, due to the variability of the search process, this fulfillment for the relative Euclidean distance must satisfy a specific number of consecutive generations. This means that the size of the initial population will be increased as long as new fitness information is provided by the last augmented population, otherwise, the population size is not increased. In general, it is expected that a smaller population size provides an unsteady fitness information, whereas a greater population size provides a steady fitness information.

The number of augmented populations required for obtaining a steady fitness distribution depends on (1) the characteristics of the optimization problem and (2) the stopping criterion defined by the EES. The suggested stopping criterion in this paper is to stop the increase of initial population when the relative Euclidean distance is less than a tolerance error ϵ in K consecutive iterations. Thus, the number of additional populations in EES is directly proportional to the K consecutive iterations and inversely proportional to the value of ϵ. Based on preliminary experimentation, it is suggested a value of 0.01 for the tolerance error ϵ, i.e., 1% of the relative Euclidean distance. Once the EES has finished, a selected subset of the initial population is provided to the EA in the next stage. For instance, if n refers to the allowed number of new individuals for each generation and a population $\mathcal{P}$ of size $m = Jn$ was generated by the EES after J iterations, the selected subset of the initial population can be done by selecting the best n individuals from the population $\mathcal{P}$. A general description of the EES is shown in Algorithm 1.

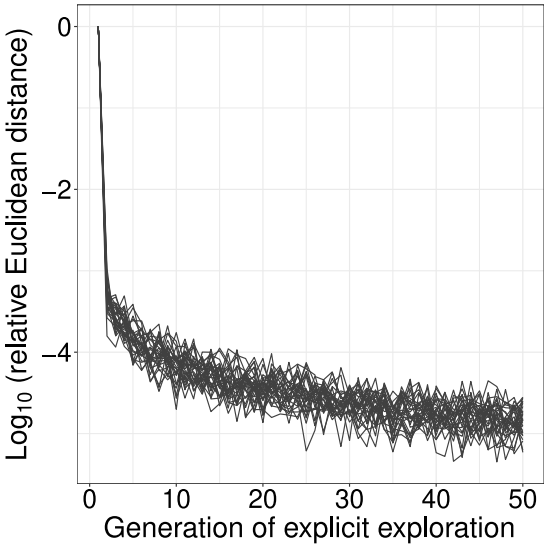


Fig. 2. Trajectories of the relative Euclidean distance for the Ackley function in the search space $[-100, 100]^{20}$.

The EES promotes the investment of an unbiased and random search in order to increase the effectiveness of the evolutionary algorithms. This strategy can decrease the risk of premature convergence and may help manage the search procedure in evolutionary algorithms. It is worthwhile to note that the EES represents a beneficial policy of a wide applicability in evolutionary algorithms. The EES takes into account the characteristics of the optimization problems for determining the size of the initial population.

3. Experiments

A set of well-known global optimization problems was chosen to test the performance of 4 evolutionary algorithms by the state-of-the-art:

Algorithm 1 Pseudocode for Explicit Exploration Strategy.

Input: Allowed number of new individuals for each generation n, fitness function $f()$, tolerance error ϵ, number of consecutive generations K.
Output: A population $\mathcal{S}$ with the n best individuals.

Generate the initial population, $\mathcal{P} \longleftarrow$ `random`(n), with n individuals at random.
Evaluate the initial population, $Y \longleftarrow f(\mathcal{P})$
Calculate deciles, $D_a \longleftarrow$ `deciles`(Y)
Set the initial error, $E \longleftarrow \epsilon + 1$
Set the initial generation, $G \longleftarrow 0$
while $G < K$ **do**
  Generate a population, $\mathcal{Q} \longleftarrow$ `random`(n)
  Evaluate population, $Z \longleftarrow f(\mathcal{Q})$
  $\mathcal{P} \longleftarrow \mathcal{P} \cup \mathcal{Q}$
  $Y \longleftarrow Y \cup Z$
  Calculate deciles, $D_b \longleftarrow$ `deciles`(Y)
  Update error, $E \longleftarrow \|D_a - D_b\|/\|D_a\|$
  if $E > \epsilon$ **then**
    $G \longleftarrow 0$
  else
    $G \longleftarrow G + 1$
  end if
  $D_a \longleftarrow D_b$
end while
Select a subset $\mathcal{S}$ from $\mathcal{P}$ according to the selection method.

- Estimation of Multivariate Normal Algorithm (EMNA).
- Differential Evolution (DE).
- Particle Swarm Optimization with Star Neighborhood (PSO-G).
- Particle Swarm Optimization with Ring Neighborhood (PSO-L).

The 4 algorithms were created to optimize continuous functions and represent the most popular and frequently used models [24]. In the case of DE algorithm the parameters used were those published by Feoktistov [25]. The parameters used for the PSO-L and PSO-G algorithms were those suggested by Clerc and Kennedy [26]. The EMNA algorithm is based on the one published by Larrañaga and Lozano [27].

In order to develop a clear analysis of the contribution of the explicit exploration strategy; only evolutionary algorithms for continuous variables were considered. Therefore, popular evolutionary algorithms, such as the genetic algorithm (originally proposed for discrete optimization), were discarded in this work.

In the first stage, each algorithm is applied to optimize the benchmark presented in Section 3.1. In the second stage, the EES is applied before executing each algorithm. The aim is not to develop a comparison among the 4 algorithms. The goal is to make a fair comparison of the algorithm without and with the EES.

For the second stage, the Algorithm 1 is applied with the following fixed parameters to execute the EES: allowed number of new individuals for each generation $n = 10D$ (D:dimension), tolerance error $\epsilon = 0.01$ and $K = 10$ consecutive generations. The chosen value for n is suitable for the experiments done in this paper (dimension ≤ 100). The value of ϵ refers to the maximum tolerance for a relative error of 1%, other values were tested, but their performances were worse. The choice of K refers to the lower bound of generations for the explicit exploration and it can be proposed as a percentage of the maximum number of generations. Given that the maximum number of generations in this work is equal to 150, the proposed value for K is equal to 10, which means that at least the first 6.7% of all generations will be related to the explicit exploration.

3.1. Global optimization benchmark

The benchmark is composed by the 10 minimization problems used in the 2019 IEEE Congress on Evolutionary Computation (CEC 2019) [28]. There are unimodal and multimodal functions. All functions are unconstrained and have a single objective. The benchmark includes classic functions used in the state-of-the-art such as the Ackley and Rastrigin function, as well as complex functions such as the Happy Cat function. The reason for choosing these global optimization problems is to understand the behavior of the 4 evolutionary algorithms in different scalable functions.

Storn's Chebyshev Polynomial Fitting Problem

$$f(\boldsymbol{x}) = p_1 + p_2 + p_3$$

$$p_1 = \begin{cases} (u-d)^2 & \text{if } u < d \\ 0 & \text{otherwise} \end{cases}$$

$$p_2 = \begin{cases} (v-d)^2 & \text{if } v < d \\ 0 & \text{otherwise} \end{cases}$$

$$p_3 = \sum_{k=0}^{32D} p_k$$

$$p_k = \begin{cases} (w_k - 1)^2 & \text{if } w_k > 1 \\ (w_k + 1)^2 & \text{if } w_k < -1 \\ 0 & \text{otherwise} \end{cases}$$

$$u = \sum_{j=1}^{D} x_j (1.2)^{D-j}$$

$$v = \sum_{j=1}^{D} x_j (-1.2)^{D-j}$$

$$w_k = \sum_{j=1}^{D} x_j \left(\frac{2k}{m} - 1\right)^{D-j}$$

$$d = \begin{cases} 72.661 & \text{if } D = 9 \\ 10558.145 & \text{if } D = 17 \end{cases}$$

$$\boldsymbol{x} \in [-8192, 8192]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Inverse Hilbert Matrix Problem

$$f(\boldsymbol{x}) = \sum_{i=1}^{n} \sum_{k=1}^{n} |w_{i,k}|$$

$$\mathbf{W} = \mathbf{HZ} - \mathbf{I}$$

$$\mathbf{W} = (w_{i,k})$$

$$\mathbf{H} = (h_{i,k})$$

$$\mathbf{Z} = (z_{i,k})$$

$$h_{i,k} = \frac{1}{i + k - 1}$$

$$z_{i,k} = x_{i+n(k-1)}$$

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$n = \sqrt{D}$$

$$i, k = 1, 2, \ldots, n$$

$$\boldsymbol{x} \in [-16384, 16384]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Lennard-Jones Minimum Energy Cluster

$$f(\boldsymbol{x}) = 12.7120622568 + \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \left(\frac{1}{d_{i,j}^2} - \frac{2}{d_{i,j}}\right)$$

$$d_{i,j} = \left(\sum_{k=0}^{2} \left(x_{3i+k-2} - x_{3j+k-2}\right)^2\right)^3$$

$$n = \frac{D}{3}$$

$$\boldsymbol{x} \in [-4, 4]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Rastrigin's Function

$$f(\boldsymbol{x}) = \sum_{i=1}^{D} \left(x_i^2 - 10cos(2\pi x_i) + 10\right)$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Griewank's Function

$$f(\boldsymbol{x}) = 1 + \frac{1}{4000}\sum_{i=1}^{D} x_i^2 - \prod_{i=1}^{D} cos\left(\frac{x_i}{\sqrt{i}}\right)$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Weierstrass Function

$$f(\boldsymbol{x}) = \sum_{i=1}^{D}\left(\sum_{k=0}^{k_{\max}}\left[a^k cos\left(2\pi b^k(x_i + 0.5)\right)\right]\right) - D\sum_{k=0}^{k_{\max}} a^k cos\left(\pi b^k\right)$$
$$a = 0.5, \qquad b = 3, \qquad k_{\max} = 20$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Modified Schwefel's Function

$$f(\boldsymbol{x}) = 418.9829D - \sum_{i=1}^{D} g(z_i)$$
$$z_i = x_i + 420.9687462275036$$
$$m_i = \text{mod}(|z_i|, 500)$$
$$g(z_i) = \begin{cases} z_i \sin(\sqrt{|z_i|}) & \text{if } |z_i| \le 500 \\ (500 - m_i)\sin(\sqrt{|500 - m_i|}) - \dfrac{(z_i - 500)^2}{10000D} & \text{if } z_i > 500 \\ (m_i - 500)\sin(\sqrt{|m_i - 500|}) - \dfrac{(z_i + 500)^2}{10000D} & \text{if } z_i < -500 \end{cases}$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Expanded Schaffer's F6 Function

$$f(\boldsymbol{x}) = g(x_1, x_2) + g(x_2, x_3) + \cdots + g(x_{D-1}, x_D) + g(x_D, x_1)$$
$$g(a, b) = 0.5 + \frac{\sin^2\left(\sqrt{a^2 + b^2}\right) - 0.5}{\left(1 + 0.001\left(a^2 + b^2\right)\right)^2}$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Happy Cat Function

$$f(\boldsymbol{x}) = \left|\sum_{i=1}^{D} x_i^2 - D\right|^{1/4} + \frac{\left(0.5\sum_{i=1}^{D} x_i^2 + \sum_{i=1}^{D} x_i\right)}{D} + 0.5$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

Ackley Function

$$f(\boldsymbol{x}) = 20 + e - 20e^{-0.2\sqrt{\frac{1}{D}\sum_{i=1}^{D} x_i^2}} - e^{\frac{1}{D}\sum_{i=1}^{D} cos(2\pi x_i)}$$
$$\boldsymbol{x} \in [-100, 100]^D, \qquad f_{\min}(\boldsymbol{x}^*) = 0$$

The Storn's Chebyshev Polynomial Fitting Problem is valid for dimensions $D = 9$ and $D = 17$. Inverse Hilbert Matrix Problem is valid for dimensions $D = n^2$, where $n = 2, 3, \ldots, 10$. The Lennard-Jones Minimum Energy Cluster function was optimized for dimensions $D = 6$ and $D = 9n$, where $n = 1, 2, \ldots, 11$. The remaining functions were optimized for dimensions $D = 2, D = 5$ and $D = 10n$, where $n = 1, 2, \ldots, 10$.

For each function and for each algorithm, 30 runs were executed. In each run 150 generations were executed. The tests were implemented in the statistical software R.

Table 1
Average relative difference in computational effort.

Function	EMNA	DE	PSO-G	PSO-L
StornChebyshev	0.64%	0.51%	0.99%	0.76%
InverseHilbert	4.34%	1.78%	8.40%	8.29%
LennardJones	1.53%	0.68%	2.02%	3.11%
Rastrigin	4.61%	4.77%	41.24%	20.96%
Griewangk	4.59%	3.53%	35.94%	19.19%
Weierstrass	0.10%	0.66%	0.00%	0.24%
ModifiedSchwefel	9.19%	1.86%	27.67%	18.94%
ExpandedSchaffer	1.58%	4.13%	11.29%	9.62%
HappyCat	5.93%	5.27%	42.07%	27.29%
Ackley	3.63%	3.01%	15.54%	11.05%

Relative difference calculated based on the execution time of the evolutionary algorithms without EES.

4. Results and discussion

The aim is to perform an unbiased study for showing the performance of EES. Thereby, each algorithm (PSO-G, PSO-L, DE and EMNA) was applied to optimize each function explained in Section 3.1 in the valid dimensions described above (In total 107 dimensions along 10 functions). The experiment was replicated by adding the EES to each algorithm. Thus, a total of 856 experiments were developed with 30 runs for each one. For every run and for each algorithm version (with and without EES), a fixed number of 150 generations were realized. The comparison between the evolutionary algorithms without EES and evolutionary algorithms with EES are based on the same number of fitness function evaluations $10 * D$, where D is the dimension tested for each function described in Section 3.1.

For example, Fig. 3 shows the evolution of the trajectory of the best fitness value for the Happy Cat function in a 50 dimensional search space across 150 generations. Gray trajectories describe the performance of EMNA algorithm for 30 runs. Black trajectories describe the performance of the EMNA+EES algorithm for 30 runs. The generations are represented in the X-axis and the fitness corresponds to the Y-axis in logarithmic scale. The first stagnation shown by the EMNA+EES during the initial generations is due to the predominance of the exploration over the exploitation. For instance, one of the black trajectories in Fig. 3 required 11 generations for the EES and the remaining 139 generations were performed by the EMNA, thus a total of 150 generations are done by EMNA+EES. In general, the EMNA trajectories descend faster than the EMNA+EES trajectories because there exist a premature exploitation stage. Nevertheless, after some generations and an adequate exploration, the EMNA+EES trajectories converge to lower values than EMNA trajectories.

Also, Fig. 3 presents the trajectory of the average fitness across 150 generations for the 30 runs explained above. It is clear that the EES improves EMNA performance.

Table 1 shows the relative difference in the computational effort of each evolutionary algorithm, on each function, as a percentage of the additional execution time that the evolutionary algorithms with EES versus without EES. For instance, the average execution time for the Ackley function in EMNA was 1.567 s and in EMNA+EES it was 1.624 s, the time difference is 0.057 s; therefore, the relative difference is $0.57/1.567 * 100\% = 3.63\%$.

4.1. Results analysis

Tables 2–11 are divided in 4 parts that correspond to the experiments developed for the 4 evolutionary algorithms. The best average fitness is highlighted in bold font for each algorithm. Below each mean, the standard deviation of the 30 runs is indicated in parentheses. The $*$ is used to represent the version

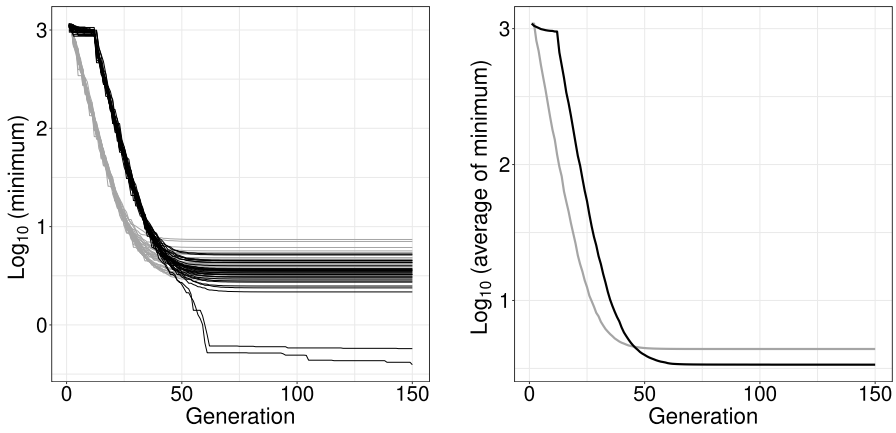


Fig. 3. Evolution of the trajectory of the best fitness value (left) and average fitness (right) for the Happy Cat function in a 50 dimensional search space. Gray trajectories for the EMNA algorithm and black trajectories for the EMNA+EES algorithm.

Table 2
Mean best results of Storn's Chebyshev Polynomial Fitting Problem.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
9	9.963e+05	[a]**7.965e+05**	2.105e+07	[a]**1.376e+07**	3.454e+03	[a]**2.233e+03**	2.855e+05	**2.079e+05**
	(1.639e+06)	(1.002e+06)	(8.608e+06)	(6.331e+06)	(8.350e+02)	(5.890e+02)	(3.262e+05)	(1.850e+05)
17	4.159e+06	**2.409e+06**	9.158e+07	[a]**5.955e+07**	2.499e+02	[a]**8.584e+01**	[a]**1.982e+06**	2.975e+06
	(3.772e+06)	(1.850e+06)	(3.270e+07)	(2.375e+07)	(1.587e+02)	(5.129e+01)	(1.303e+06)	(2.145e+06)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

Table 3
Mean best results of Inverse Hilbert Matrix Problem.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
4	2.357e+01	**1.673e+01**	**1.004e+02**	1.035e+02	1.271e+00	[a]**1.009e+00**	**4.570e−01**	6.892e−01
	(4.429e+01)	(5.865e+01)	(3.906e+01)	(3.723e+01)	(2.684e−01)	(2.935e−01)	(5.766e−01)	(9.196e−01)
9	3.228e+01	**2.434e+01**	8.402e+02	[a]**7.233e+02**	2.923e+00	[a]**2.787e+00**	**7.385e+01**	7.701e+01
	(3.326e+01)	(2.125e+01)	(1.416e+02)	(1.514e+02)	(1.086e−01)	(1.763e−01)	(3.882e+01)	(3.446e+01)
16	4.280e+01	**3.975e+01**	2.768e+03	[a]**2.555e+03**	**4.000e+00**	3.043e+02	4.494e+02	[a]**3.607e+02**
	(2.593e+01)	(2.799e+01)	(3.801e+02)	(3.743e+02)	(3.706e−15)	(1.645e+03)	(1.568e+02)	(1.314e+02)
25	**4.279e+01**	4.753e+01	6.323e+03	[a]**5.530e+03**	[a]**5.000e+00**	5.000e+00	1.453e+03	[a]**1.098e+03**
	(3.883e+01)	(4.210e+01)	(9.803e+02)	(6.171e+02)	(1.814e−15)	(1.525e−13)	(4.845e+02)	(2.753e+02)
36	4.971e+01	[a]**3.238e+01**	1.135e+04	[a]**9.511e+03**	[a]**6.000e+00**	6.000e+00	2.743e+03	[a]**2.222e+03**
	(4.437e+01)	(2.437e+01)	(1.114e+03)	(1.022e+03)	(0.000e+00)	(4.853e−14)	(5.491e+02)	(4.696e+02)
49	4.700e+01	**4.062e+01**	1.756e+04	[a]**1.512e+04**	[a]**7.000e+00**	7.000e+00	4.262e+03	[a]**3.382e+03**
	(3.466e+01)	(3.609e+01)	(1.232e+03)	(1.282e+03)	(0.000e+00)	(6.243e−14)	(7.396e+02)	(5.912e+02)
64	4.703e+01	**3.623e+01**	2.461e+04	[a]**2.056e+04**	[a]**8.000e+00**	8.000e+00	6.363e+03	[a]**5.471e+03**
	(3.796e+01)	(1.325e+01)	(2.248e+03)	(2.765e+03)	(1.979e−15)	(2.335e−14)	(8.706e+02)	(8.463e+02)
81	5.210e+01	**4.550e+01**	3.221e+04	[a]**2.923e+04**	[a]**9.000e+00**	9.000e+00	9.183e+03	[a]**7.595e+03**
	(2.034e+01)	(2.183e+01)	(2.601e+03)	(2.348e+03)	(0.000e+00)	(1.705e−14)	(1.229e+03)	(7.561e+02)
100	4.750e+01	**4.576e+01**	4.176e+04	[a]**3.728e+04**	[a]**1.000e+01**	1.000e+01	1.204e+04	[a]**9.943e+03**
	(1.843e+01)	(1.840e+01)	(3.412e+03)	(2.904e+03)	(0.000e+00)	(4.637e−14)	(1.455e+03)	(1.181e+03)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

of each algorithm that is statistically better according to the Wilcoxon signed-rank test.

Table 2 presents the mean best fitness value of the 30 runs of each algorithm for the Storn's Chebyshev Polynomial Fitting Problem in dimensions 9 and 17. In general, the results obtained for this function represents a very clear advantage when applying the EES version, except in dimension 17 where the PSO-L without EES outperforms the PSO.L+EES version. The algorithms+EES denote the lowest variability, except in dimension 17 where the PSO-L without EES has a lower standard deviation than the PSO.L+EES.

Table 3 presents the mean best fitness value of the 30 runs of each algorithm for the Inverse Hilbert Matrix Problem in dimensions from 4 to 100. The table shows that for DE and PSO-L, the results obtained by the evolutionary algorithms applying EES are statistically better. In the case of the PSO-G, there is a statistical difference in dimensions from 25 to 100. The PSO.L+EES denotes less variability than the PSO-L without EES. On the other hand, the PSO-G+EES has a higher standard deviation than the PSO-G without EES. No trend can be observed in the standard deviation of the DE and EMNA algorithms.

Table 4 presents the mean best fitness value of the 30 runs of each algorithm for the Lennard-Jones Minimum Energy Cluster in dimensions from 6 to 99. The results obtained by the DE algorithm improve with the explicit exploration. In the case of the PSO-G and PSO-L, there is not an improvement with the EES version; both algorithms without EES are statistically better.

Table 4
Mean best results of Lennard-Jones Minimum Energy Cluster.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
6	−2.220e−16 (0.000e+00)	−2.220e−16 (0.000e+00)	2.576e−06 (3.877e−06)	**2.368e−06** (3.164e−06)	**6.669e−07** (1.814e−06)	7.153e−07 (9.344e−07)	8.142e−17 (1.009e−15)	**−1.221e−16** (2.265e−16)
9	1.446e−02 (4.107e−02)	**7.210e−03** (1.397e−02)	9.764e−02 (6.867e−02)	**9.134e−02** (6.666e−02)	[a]**1.078e−01** (8.785e−02)	7.061e−01 (3.559e−01)	1.897e−03 (6.972e−03)	**8.029e−04** (2.248e−03)
18	**5.829e+00** (6.453e−01)	6.016e+00 (7.498e−01)	6.861e+00 (8.287e−01)	**6.779e+00** (5.414e−01)	[a]**5.878e+00** (9.528e−01)	9.284e+00 (6.191e−01)	[a]**5.972e+00** (8.344e−01)	6.499e+00 (7.274e−01)
27	**1.538e+01** (1.135e+00)	1.578e+01 (8.729e−01)	1.681e+01 (7.793e−01)	[a]**1.646e+01** (6.199e−01)	[a]**1.515e+01** (1.406e+00)	1.913e+01 (5.504e−01)	[a]**1.499e+01** (1.142e+00)	1.622e+01 (8.771e−01)
36	[a]**2.761e+01** (1.439e+00)	2.877e+01 (1.175e+00)	2.896e+01 (7.765e−01)	**2.867e+01** (1.003e+00)	[a]**2.678e+01** (1.689e+00)	3.113e+01 (1.080e+00)	[a]**2.725e+01** (1.363e+00)	2.844e+01 (1.419e+00)
45	[a]**4.222e+01** (1.187e+00)	4.318e+01 (1.048e+00)	4.168e+01 (9.419e−01)	[a]**4.102e+01** (8.285e−01)	[a]**4.036e+01** (1.659e+00)	4.401e+01 (8.014e−01)	[a]**4.055e+01** (1.045e+00)	4.119e+01 (1.214e+00)
54	**5.644e+01** (1.014e+00)	5.672e+01 (1.125e+00)	5.408e+01 (1.125e+00)	**5.371e+01** (7.705e−01)	[a]**5.314e+01** (2.696e+00)	5.647e+01 (8.939e−01)	**5.346e+01** (1.443e+00)	5.384e+01 (1.722e+00)
63	7.130e+01 (1.397e+00)	**7.100e+01** (1.350e+00)	6.732e+01 (9.209e−01)	[a]**6.658e+01** (1.293e+00)	[a]**6.637e+01** (2.368e+00)	6.975e+01 (1.316e+00)	[a]**6.701e+01** (1.685e+00)	6.825e+01 (1.491e+00)
72	8.614e+01 (7.248e−01)	**8.607e+01** (1.304e+00)	8.134e+01 (1.072e+00)	[a]**8.070e+01** (7.387e−01)	[a]**8.163e+01** (2.222e+00)	8.373e+01 (1.478e+00)	**8.191e+01** (2.151e+00)	8.211e+01 (1.887e+00)
81	1.009e+02 (1.025e+00)	**1.005e+02** (1.466e+00)	9.523e+01 (7.833e−01)	[a]**9.401e+01** (1.080e+00)	**9.707e+01** (3.075e+00)	9.755e+01 (1.365e+00)	9.651e+01 (1.914e+00)	**9.627e+01** (2.709e+00)
90	1.152e+02 (1.607e+00)	[a]**1.145e+02** (1.438e+00)	1.083e+02 (1.196e+00)	**1.079e+02** (9.627e−01)	**1.110e+02** (3.534e+00)	1.111e+02 (1.680e+00)	**1.110e+02** (1.997e+00)	1.112e+02 (2.144e+00)
99	1.311e+02 (1.050e+00)	[a]**1.301e+02** (1.416e+00)	1.233e+02 (1.182e+00)	[a]**1.220e+02** (1.507e+00)	1.284e+02 (2.310e+00)	[a]**1.257e+02** (1.226e+00)	1.272e+02 (2.174e+00)	[a]**1.261e+02** (1.694e+00)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

Table 5
Mean best results of Rastrigin Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	5.910e−01 (6.472e−01)	**3.606e−01** (5.378e−01)	**0.000e+00** (0.000e+00)	2.072e−15 (6.952e−15)	1.327e−01 (5.049e−01)	**1.327e−01** (5.049e−01)	[a]**3.194e−02** (7.916e−02)	1.369e−01 (2.418e−01)
5	9.687e+00 (1.847e+01)	[a]**3.722e+00** (2.188e+00)	[a]**1.492e−02** (2.225e−02)	1.089e−01 (1.896e−01)	**4.975e−01** (1.518e+00)	9.950e−01 (2.024e+00)	**7.282e+00** (3.642e+00)	8.226e+00 (2.895e+00)
10	**1.648e+01** (7.950e+00)	1.822e+01 (6.705e+00)	**2.059e+01** (3.416e+00)	2.091e+01 (3.618e+00)	**3.317e−01** (1.817e+00)	1.327e+00 (3.440e+00)	9.018e+01 (3.132e+01)	[a]**7.369e+01** (1.898e+01)
20	6.516e+01 (2.973e+01)	**6.142e+01** (2.086e+01)	3.244e+02 (3.603e+01)	[a]**3.011e+02** (3.296e+01)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	4.306e+02 (9.392e+01)	[a]**3.539e+02** (7.296e+01)
30	1.365e+02 (4.851e+01)	**1.202e+02** (2.605e+01)	1.970e+03 (2.139e+02)	[a]**.734e+03** (2.280e+02)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	8.644e+02 (9.944e+01)	[a]**7.211e+02** (8.438e+01)
40	1.998e+02 (4.354e+01)	[a]**1.763e+02** (2.972e+01)	6.381e+03 (7.985e+02)	[a]**5.633e+03** (5.782e+02)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	1.346e+03 (1.757e+02)	[a]**1.165e+03** (1.421e+02)
50	2.851e+02 (5.791e+01)	[a]**2.512e+02** (4.778e+01)	1.421e+04 (1.375e+03)	[a]**1.256e+04** (9.000e+02)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	1.804e+03 (1.898e+02)	[a]**1.623e+03** (1.665e+02)
60	3.825e+02 (9.356e+01)	[a]**3.321e+02** (6.021e+01)	2.605e+04 (1.563e+03)	[a]**2.193e+04** (1.758e+03)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	2.319e+03 (2.032e+02)	[a]**1.981e+03** (1.510e+02)
70	**4.302e+02** (6.662e+01)	4.378e+02 (6.042e+01)	3.931e+04 (2.538e+03)	[a]**3.344e+04** (2.326e+03)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	2.774e+03 (2.591e+02)	[a]**2.579e+03** (1.695e+02)
80	5.205e+02 (6.805e+01)	**5.040e+02** (7.417e+01)	5.437e+04 (3.979e+03)	[a]**4.778e+04** (2.549e+03)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	3.359e+03 (2.550e+02)	[a]**3.046e+03** (2.558e+02)
90	6.288e+02 (7.531e+01)	[a]**5.814e+02** (6.665e+01)	7.516e+04 (3.558e+03)	[a]**6.424e+04** (3.589e+03)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	3.770e+03 (2.705e+02)	[a]**3.490e+03** (2.887e+02)
100	7.090e+02 (7.400e+01)	[a]**6.693e+02** (7.380e+01)	9.554e+04 (3.938e+03)	[a]**8.238e+04** (4.269e+03)	0.000e+00 (0.000e+00)	0.000e+00 (0.000e+00)	4.352e+03 (2.911e+02)	[a]**3.879e+03** (3.071e+02)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

For the EMNA algorithm, the EES gives better results in higher dimensions (63 to 99). In general, a lower standard deviation is observed in the 4 algorithms with EES.

Table 5 presents the mean best fitness value of the 30 runs of each algorithm for the Rastrigin function in dimensions from 2 to 100. For DE and PSO-L, the results obtained with the EES are statistically better. In half of the tested dimensions, the EMNA+EES obtains the best statistical results. In the case of the PSO-G, there is not a difference in dimensions from 20 to 100. The EMNA+EES and PSO.L+EES algorithms denote less variability than the EMNA and PSO.L without EES. On the other hand, the PSO-G with and without EES show a standard deviation equal to 0 in dimensions from 20 to 100.

Table 6 presents the mean best fitness value of the 30 runs of each algorithm for the Griewangk Function in dimensions from 2 to 100. For the PSO-G algorithm, the EES does not represent a difference in dimensions from 10 to 100. For DE and PSO-L, the results obtained with the EES are statistically better. In half of the tested dimensions, the EMNA+EES obtains the best statistical results; however, the EMNA+EES shows more variability than the EMNA without EES. On the other hand, the DE+EES and PSO.L+EES algorithms denote less variability than the DE and PSO.L without

Table 6
Mean best results of Griewangk Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	**1.738e−02**	1.849e−02	2.523e−03	**2.356e−03**	**8.306e−04**	1.269e−03	**1.485e−02**	1.581e−02
	(1.355e−02)	(1.716e−02)	(3.481e−03)	(3.247e−03)	(2.538e−03)	(2.887e−03)	(7.550e−03)	(8.221e−03)
5	6.018e−02	**5.258e−02**	**3.498e−02**	4.073e−02	3.631e−10	**0.000e+00**	**1.016e−01**	1.130e−01
	(5.569e−02)	(6.115e−02)	(1.302e−02)	(1.410e−02)	(1.989e−09)	(0.000e+00)	(5.409e−02)	(5.407e−02)
10	2.551e−01	**1.921e−01**	2.099e−01	**1.899e−01**	0.000e+00	0.000e+00	4.570e−01	[a]**3.913e−01**
	(2.151e−01)	(1.838e−01)	(7.214e−02)	(6.106e−02)	(0.000e+00)	(0.000e+00)	(1.250e−01)	(1.073e−01)
20	5.167e−01	[a]**3.821e−01**	1.031e+00	[a]**1.022e+00**	0.000e+00	0.000e+00	1.058e+00	[a]**1.047e+00**
	(2.846e−01)	(2.416e−01)	(1.171e−02)	(1.594e−02)	(0.000e+00)	(0.000e+00)	(2.605e−02)	(2.160e−02)
30	7.211e−01	**6.684e−01**	1.406e+00	[a]**1.356e+00**	0.000e+00	0.000e+00	1.154e+00	[a]**1.112e+00**
	(2.007e−01)	(2.185e−01)	(5.920e−02)	(5.473e−02)	(0.000e+00)	(0.000e+00)	(2.698e−02)	(2.100e−02)
40	8.784e−01	[a]**7.934e−01**	2.513e+00	[a]**2.304e+00**	0.000e+00	0.000e+00	1.233e+00	[a]**1.185e+00**
	(1.524e−01)	(2.082e−01)	(1.819e−01)	(1.259e−01)	(0.000e+00)	(0.000e+00)	(3.817e−02)	(3.022e−02)
50	9.629e−01	[a]**8.249e−01**	4.449e+00	[a]**4.010e+00**	0.000e+00	0.000e+00	1.336e+00	[a]**1.282e+00**
	(7.662e−02)	(1.737e−01)	(3.248e−01)	(1.992e−01)	(0.000e+00)	(0.000e+00)	(5.390e−02)	(3.954e−02)
60	9.986e−01	[a]**8.825e−01**	7.247e+00	[a]**6.267e+00**	0.000e+00	0.000e+00	1.438e+00	[a]**1.372e+00**
	(3.954e−02)	(1.035e−01)	(4.623e−01)	(4.275e−01)	(0.000e+00)	(0.000e+00)	(3.821e−02)	(4.096e−02)
70	9.997e−01	**9.640e−01**	1.071e+01	[a]**9.316e+00**	0.000e+00	0.000e+00	1.537e+00	[a]**1.437e+00**
	(9.867e−02)	(7.452e−02)	(6.971e−01)	(4.928e−01)	(0.000e+00)	(0.000e+00)	(5.070e−02)	(4.806e−02)
80	1.016e+00	**9.933e−01**	1.507e+01	[a]**1.298e+01**	0.000e+00	0.000e+00	1.654e+00	[a]**1.554e+00**
	(5.799e−02)	(6.626e−02)	(7.544e−01)	(8.004e−01)	(0.000e+00)	(0.000e+00)	(6.426e−02)	(6.917e−02)
90	1.042e+00	[a]**1.011e+00**	1.911e+01	[a]**1.693e+01**	0.000e+00	0.000e+00	1.733e+00	[a]**1.649e+00**
	(4.644e−02)	(7.247e−02)	(9.580e−01)	(7.711e−01)	(0.000e+00)	(0.000e+00)	(7.387e−02)	(6.101e−02)
100	1.059e+00	[a]**1.021e+00**	2.405e+01	[a]**2.137e+01**	0.000e+00	0.000e+00	1.852e+00	[a]**1.750e+00**
	(3.749e−02)	(5.557e−02)	(1.661e+00)	(7.674e−01)	(0.000e+00)	(0.000e+00)	(6.810e−02)	(6.419e−02)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

Table 7
Mean best results of Weierstrass Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	1.722e−01	**1.364e−01**	**1.601e−02**	1.791e−02	[a]**4.313e−03**	1.288e−01	4.438e−01	**4.144e−01**
	(2.096e−01)	(1.901e−01)	(8.370e−03)	(8.437e−03)	(1.624e−02)	(1.452e−01)	(1.357e−01)	(1.622e−01)
5	3.091e+00	**2.988e+00**	7.250e−01	**7.127e−01**	[a]**6.116e−08**	5.974e−05	**3.183e+00**	3.227e+00
	(6.567e−01)	(6.714e−01)	(1.307e−01)	(1.521e−01)	(3.309e−07)	(1.335e−04)	(4.262e−01)	(5.050e−01)
10	9.992e+00	**9.955e+00**	4.243e+00	**4.086e+00**	**1.057e−12**	2.550e−07	9.934e+00	**9.837e+00**
	(7.106e−01)	(6.855e−01)	(4.176e−01)	(4.653e−01)	(4.638e−12)	(1.392e−06)	(9.862e−01)	(7.230e−01)
20	2.499e+01	**2.495e+01**	**1.521e+01**	1.552e+01	**7.958e−14**	3.691e−10	2.526e+01	**2.513e+01**
	(1.002e+00)	(1.104e+00)	(1.093e+00)	(7.914e−01)	(2.316e−13)	(1.960e−09)	(7.871e−01)	(1.048e+00)
30	**4.144e+01**	4.145e+01	2.854e+01	**2.810e+01**	**1.120e−13**	2.661e−05	4.172e+01	**4.147e+01**
	(1.619e+00)	(1.021e+00)	(1.079e+00)	(1.317e+00)	(1.806e−13)	(1.154e−04)	(1.015e+00)	(1.244e+00)
40	**5.812e+01**	5.846e+01	**4.217e+01**	4.253e+01	**1.378e−13**	3.305e−11	5.826e+01	**5.801e+01**
	(1.288e+00)	(1.015e+00)	(1.058e+00)	(1.199e+00)	(2.368e−13)	(1.532e−10)	(1.388e+00)	(1.648e+00)
50	7.514e+01	**7.494e+01**	**5.641e+01**	5.642e+01	**1.099e−13**	7.080e−11	7.526e+01	**7.526e+01**
	(1.686e+00)	(1.684e+00)	(1.413e+00)	(1.502e+00)	(2.136e−13)	(3.861e−10)	(1.349e+00)	(1.612e+00)
60	9.318e+01	[a]**9.223e+01**	7.113e+01	[a]**7.020e+01**	**1.061e−13**	5.304e−08	**9.155e+01**	9.226e+01
	(1.155e+00)	(1.516e+00)	(1.578e+00)	(2.065e+00)	(2.295e−13)	(2.818e−07)	(1.962e+00)	(1.751e+00)
70	**1.100e+02**	1.103e+02	**8.579e+01**	8.591e+01	**3.562e−13**	2.868e−08	**1.098e+02**	1.099e+02
	(1.615e+00)	(1.603e+00)	(1.419e+00)	(1.329e+00)	(7.281e−13)	(1.568e−07)	(1.694e+00)	(1.729e+00)
80	1.281e+02	[a]**1.274e+02**	1.015e+02	[a]**1.006e+02**	[a]**2.128e−12**	2.537e−11	[a]**1.274e+02**	1.282e+02
	(1.666e+00)	(1.544e+00)	(1.412e+00)	(1.469e+00)	(1.075e−11)	(7.281e−11)	(2.180e+00)	(1.532e+00)
90	**1.452e+02**	1.461e+02	1.169e+02	**1.165e+02**	**2.937e−13**	1.963e−08	**1.453e+02**	1.461e+02
	(2.292e+00)	(1.994e+00)	(1.464e+00)	(1.750e+00)	(6.701e−13)	(1.072e−07)	(2.350e+00)	(1.824e+00)
100	1.637e+02	**1.635e+02**	1.322e+02	**1.321e+02**	**1.052e−13**	3.177e−11	**1.634e+02**	1.638e+02
	(3.025e+00)	(1.891e+00)	(2.064e+00)	(2.180e+00)	(2.314e−13)	(1.726e−10)	(2.355e+00)	(2.382e+00)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

EES. Finally, the PSO-G with and without EES show a standard deviation equal to 0 in dimensions from 10 to 100.

Table 7 presents the mean best fitness value of the 30 runs of each algorithm for the Weierstrass Function in dimensions from 2 to 100. The table shows that for the PSO-G algorithm the results are worse with the EES in all dimensions. There is not a conclusion, in the case of the EMNA, DE and PSO-L algorithms. Nevertheless, the variability of the DE+EES, PSO-G+EES, and PSO-L+EES algorithms is greater than the DE, PSO-G, and PSO-L without EES. Note that in the case of PSO-G versus PSO-G+EES, the difference is overwhelming; nevertheless, the PSO-G is statistically better only in 3 dimensions.

Table 8 presents the mean best fitness value of the 30 runs of each algorithm for the Modified Schwefel's Function in dimensions from 2 to 100. The table shows that for EMNA, DE and PSO-L, the EES improves the results obtained by the evolutionary algorithms. In the case of the PSO-G, there is not a difference when applying explicit exploration. The standard deviation of EMNA+EES, DE+EES, and PSO-G+EES is lower than the 3 algorithms without EES; being a very clear difference in small dimensions.

Table 9 presents the mean best fitness value of the 30 runs of each algorithm for the Expanded Schaffer's F6 Function in dimensions from 2 to 100. The table shows that the DE+EES is

Table 8
Mean best results of Modified Schwefel's Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	9.106e−02	**8.319e−03**	2.546e−05	2.546e−05	2.546e−05	2.546e−05	[a]**2.546e−05**	2.546e−05
	(4.310e−01)	(4.508e−02)	(0.000e+00)	(0.000e+00)	(0.000e+00)	(0.000e+00)	(4.577e−14)	(1.008e−13)
5	6.843e−01	[a]**7.922e−02**	6.370e−05	[a]**6.365e−05**	6.364e−05	6.364e−05	1.822e−02	[a]**1.541e−03**
	(1.328e+00)	(2.047e−01)	(1.047e−07)	(1.252e−08)	(0.000e+00)	(0.000e+00)	(5.321e−02)	(3.955e−03)
10	1.114e+00	**4.622e−01**	5.131e−01	[a]**3.074e−02**	1.273e−04	1.273e−04	3.007e+00	[a]**1.892e+00**
	(2.028e+00)	(7.826e−01)	(4.082e−01)	(1.575e−02)	(4.239e−13)	(4.458e−13)	(2.014e+00)	(1.208e+00)
20	5.753e+00	[a]**2.819e+00**	1.798e+02	[a]**6.501e+01**	2.546e−04	2.546e−04	3.556e+01	[a]**2.494e+01**
	(5.399e+00)	(2.503e+00)	(3.629e+01)	(1.796e+01)	(0.000e+00)	(0.000e+00)	(1.212e+01)	(8.821e+00)
30	1.111e+01	[a]**6.918e+00**	8.341e+02	[a]**5.070e+02**	3.818e−04	3.818e−04	7.489e+01	[a]**6.085e+01**
	(7.166e+00)	(3.977e+00)	(1.034e+02)	(6.005e+01)	(0.000e+00)	(0.000e+00)	(1.596e+01)	(1.139e+01)
40	1.640e+01	[a]**9.506e+00**	1.815e+03	[a]**1.325e+03**	5.091e−04	5.091e−04	1.275e+02	[a]**9.933e+01**
	(7.664e+00)	(5.313e+00)	(1.791e+02)	(1.110e+02)	(0.000e+00)	(0.000e+00)	(2.123e+01)	(1.642e+01)
50	2.194e+01	[a]**1.553e+01**	3.033e+03	[a]**2.220e+03**	6.364e−04	6.364e−04	1.792e+02	[a]**1.516e+02**
	(9.038e+00)	(6.255e+00)	(2.292e+02)	(2.326e+02)	(0.000e+00)	(0.000e+00)	(2.316e+01)	(2.000e+01)
60	3.300e+01	[a]**2.310e+01**	4.258e+03	[a]**3.497e+03**	7.637e−04	7.637e−04	2.217e+02	[a]**1.990e+02**
	(1.609e+01)	(9.830e+00)	(2.435e+02)	(2.067e+02)	(0.000e+00)	(6.642e−13)	(2.576e+01)	(1.846e+01)
70	4.390e+01	[a]**2.717e+01**	5.605e+03	[a]**4.679e+03**	8.909e−04	8.909e−04	2.692e+02	[a]**2.380e+02**
	(1.489e+01)	(8.487e+00)	(3.828e+02)	(2.760e+02)	(6.642e−13)	(0.000e+00)	(2.684e+01)	(2.662e+01)
80	4.896e+01	[a]**3.527e+01**	7.044e+03	[a]**6.116e+03**	1.018e−03	1.018e−03	3.270e+02	[a]**2.860e+02**
	(1.333e+01)	(9.611e+00)	(3.579e+02)	(3.664e+02)	(0.000e+00)	(0.000e+00)	(2.564e+01)	(3.330e+01)
90	5.479e+01	[a]**3.960e+01**	8.650e+03	[a]**7.580e+03**	1.145e−03	1.145e−03	3.749e+02	[a]**3.411e+02**
	(1.735e+01)	(9.155e+00)	(3.356e+02)	(2.646e+02)	(0.000e+00)	(0.000e+00)	(3.128e+01)	(3.172e+01)
100	5.986e+01	[a]**4.919e+01**	1.019e+04	[a]**9.230e+03**	1.273e−03	1.273e−03	4.406e+02	[a]**3.839e+02**
	(1.482e+01)	(1.274e+01)	(3.552e+02)	(2.542e+02)	(0.000e+00)	(0.000e+00)	(3.470e+01)	(2.576e+01)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

Table 9
Mean best results of Expanded Schaffer's F6 Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	3.008e−02	**2.245e−02**	1.359e−02	**1.291e−02**	5.182e−03	**4.534e−03**	1.050e−01	[a]**2.842e−02**
	(3.116e−02)	(1.457e−02)	(7.395e−03)	(7.638e−03)	(8.740e−03)	(8.359e−03)	(7.603e−02)	(1.708e−02)
5	9.556e−01	**9.054e−01**	2.033e−01	**1.874e−01**	[a]**3.184e−02**	8.824e−02	1.541e+00	[a]**1.330e+00**
	(2.257e−01)	(3.045e−01)	(6.490e−02)	(4.420e−02)	(4.813e−02)	(1.483e−01)	(2.280e−01)	(2.123e−01)
10	2.914e+00	**2.847e+00**	1.581e+00	[a]**1.287e+00**	[a]**1.619e−02**	1.067e−01	3.701e+00	[a]**3.596e+00**
	(3.995e−01)	(3.766e−01)	(2.361e−01)	(2.692e−01)	(3.683e−02)	(2.689e−01)	(1.680e−01)	(2.360e−01)
20	**7.337e+00**	7.372e+00	6.670e+00	[a]**6.239e+00**	**3.886e−02**	6.368e−02	8.102e+00	**8.080e+00**
	(3.602e−01)	(4.489e−01)	(2.371e−01)	(2.670e−01)	(7.906e−02)	(1.508e−01)	(2.238e−01)	(2.547e−01)
30	1.187e+01	**1.183e+01**	1.160e+01	[a]**1.105e+01**	**9.716e−03**	9.716e−03	[a]**1.261e+01**	1.275e+01
	(4.032e−01)	(4.060e−01)	(2.531e−01)	(4.055e−01)	(5.322e−02)	(5.322e−02)	(3.392e−01)	(1.879e−01)
40	**1.652e+01**	1.653e+01	1.632e+01	[a]**1.575e+01**	**0.000e+00**	2.591e−02	**1.726e+01**	1.730e+01
	(4.889e−01)	(4.635e−01)	(3.126e−01)	(2.106e−01)	(0.000e+00)	(9.860e−02)	(3.861e−01)	(3.676e−01)
50	2.113e+01	**2.110e+01**	2.096e+01	[a]**2.045e+01**	0.000e+00	0.000e+00	**2.196e+01**	2.197e+01
	(4.405e−01)	(4.444e−01)	(2.766e−01)	(3.000e−01)	(0.000e+00)	(0.000e+00)	(4.577e−01)	(3.767e−01)
60	**2.585e+01**	2.585e+01	2.572e+01	[a]**2.521e+01**	0.000e+00	0.000e+00	2.672e+01	**2.669e+01**
	(5.156e−01)	(5.814e−01)	(3.359e−01)	(2.359e−01)	(0.000e+00)	(0.000e+00)	(3.222e−01)	(4.301e−01)
70	3.051e+01	**3.042e+01**	3.043e+01	[a]**2.974e+01**	0.000e+00	0.000e+00	**3.134e+01**	3.140e+01
	(5.924e−01)	(6.326e−01)	(3.860e−01)	(3.638e−01)	(0.000e+00)	(0.000e+00)	(5.307e−01)	(3.737e−01)
80	**3.521e+01**	3.524e+01	3.521e+01	[a]**3.455e+01**	0.000e+00	0.000e+00	3.595e+01	**3.594e+01**
	(4.904e−01)	(6.929e−01)	(2.399e−01)	(2.809e−01)	(0.000e+00)	(0.000e+00)	(5.430e−01)	(5.228e−01)
90	**4.019e+01**	4.020e+01	4.003e+01	[a]**3.933e+01**	0.000e+00	0.000e+00	4.083e+01	**4.076e+01**
	(5.528e−01)	(5.643e−01)	(2.144e−01)	(2.707e−01)	(0.000e+00)	(0.000e+00)	(4.782e−01)	(4.360e−01)
100	**4.482e+01**	4.485e+01	4.462e+01	[a]**4.405e+01**	0.000e+00	0.000e+00	**4.561e+01**	4.573e+01
	(5.706e−01)	(5.055e−01)	(3.500e−01)	(3.168e−01)	(0.000e+00)	(0.000e+00)	(5.052e−01)	(4.352e−01)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

statistically better. The results obtained in the EMNA and PSO-L algorithms are not conclusive. In the case of the PSO-G VS PSO-G+EES, both algorithms obtained the same results for dimensions from 50 to 100. The PSO-L+EES algorithm has less variability than the PSO-L algorithm. No difference in standard deviation can be observed for EMNA, DE, and PSO-G with and without EES.

Table 10 presents the mean best fitness value of the 30 runs of each algorithm for the Happy Cat Function in dimensions from 2 to 100. The table shows that for EMNA, DE and PSO-L, the results obtained with the EES are statistically better. In the case of the PSO-G, there is not a conclusion. On the other hand, the EMNA+EES and PSO-L+EES algorithms denote less standard deviation than the EMNA and PSO-L without EES.

Table 11 presents the mean best fitness value of the 30 runs of each algorithm for the Ackley Function in dimensions from 2 to 100. The table shows that for the PSO-G vs PSO-G+EES algorithms, but there is not a difference in dimensions from 20 to 100. For the PSO-L algorithm, the EES worsen the results in dimensions from 30 to 100. In the case of EMNA and DE, the results obtained by the evolutionary algorithms with and without EES are not conclusive. In relation to the standard deviation, the EMNA and PSO-L algorithms show greater variability than EMNA+ESS and PSO-L+EES, respectively.

Table 10
Mean best results of Happy Cat Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	1.634e−01	**4.090e−02**	**1.382e−01**	1.564e−01	**6.900e−02**	7.506e−02	**1.589e−02**	1.823e−02
	(4.871e−01)	(9.933e−02)	(4.005e−02)	(4.134e−02)	(3.267e−02)	(3.896e−02)	(1.226e−02)	(1.255e−02)
5	5.488e−01	**5.308e−02**	**2.186e−01**	2.308e−01	**5.539e−02**	6.555e−02	**1.422e−01**	1.663e−01
	(1.851e+00)	(5.738e−02)	(5.370e−02)	(4.320e−02)	(2.934e−02)	(4.006e−02)	(9.479e−02)	(7.430e−02)
10	8.329e−01	[a]**3.021e−01**	3.958e−01	**3.837e−01**	6.347e−02	**5.127e−02**	2.970e+00	[a]**2.343e+00**
	(1.437e+00)	(6.632e−01)	(7.326e−02)	(7.309e−02)	(3.066e−02)	(2.598e−02)	(1.610e+00)	(1.180e+00)
20	1.500e+00	**1.093e+00**	7.025e+00	[a]**6.515e+00**	**5.002e−02**	5.302e−02	1.137e+01	[a]**8.247e+00**
	(1.398e+00)	(1.104e+00)	(9.781e−01)	(8.369e−01)	(2.063e−02)	(3.191e−02)	(2.448e+00)	(1.609e+00)
30	2.900e+00	**2.380e+00**	3.307e+01	[a]**2.989e+01**	**4.859e−02**	5.080e−02	1.435e+01	[a]**1.291e+01**
	(1.713e+00)	(1.318e+00)	(4.642e+00)	(3.727e+00)	(2.503e−02)	(3.082e−02)	(2.253e+00)	(1.842e+00)
40	3.823e+00	[a]**3.005e+00**	8.417e+01	[a]**7.430e+01**	5.532e−02	[a]**4.400e−02**	1.758e+01	[a]**1.464e+01**
	(1.200e+00)	(1.082e+00)	(1.011e+01)	(8.647e+00)	(2.857e−02)	(1.834e−02)	(1.881e+00)	(1.806e+00)
50	4.396e+00	[a]**3.370e+00**	1.525e+02	[a]**1.293e+02**	**4.221e−02**	4.257e−02	1.937e+01	[a]**1.702e+01**
	(1.165e+00)	(1.088e+00)	(8.734e+00)	(9.158e+00)	(1.939e−02)	(2.679e−02)	(1.844e+00)	(1.463e+00)
60	4.632e+00	[a]**4.020e+00**	2.142e+02	[a]**1.926e+02**	4.929e−02	**4.695e−02**	2.120e+01	[a]**1.879e+01**
	(9.584e−01)	(8.260e−01)	(1.480e+01)	(1.463e+01)	(2.117e−02)	(2.983e−02)	(1.822e+00)	(1.399e+00)
70	5.019e+00	[a]**4.452e+00**	2.874e+02	[a]**2.561e+02**	4.808e−02	**4.740e−02**	2.201e+01	[a]**1.973e+01**
	(9.760e−01)	(5.980e−01)	(1.554e+01)	(1.400e+01)	(2.399e−02)	(2.267e−02)	(1.768e+00)	(1.550e+00)
80	5.257e+00	[a]**4.573e+00**	3.628e+02	[a]**3.084e+02**	5.006e−02	**4.505e−02**	2.326e+01	[a]**2.045e+01**
	(7.754e−01)	(6.593e−01)	(2.198e+01)	(1.927e+01)	(2.372e−02)	(2.269e−02)	(1.397e+00)	(1.806e+00)
90	5.270e+00	[a]**4.952e+00**	4.263e+02	[a]**3.672e+02**	**4.764e−02**	5.811e−02	2.401e+01	[a]**2.164e+01**
	(6.493e−01)	(7.231e−01)	(1.914e+01)	(2.283e+01)	(2.602e−02)	(2.535e−02)	(1.494e+00)	(1.258e+00)
100	5.424e+00	**5.322e+00**	4.792e+02	[a]**4.235e+02**	**4.500e−02**	4.842e−02	2.450e+01	[a]**2.232e+01**
	(6.691e−01)	(7.780e−01)	(2.680e+01)	(1.987e+01)	(1.845e−02)	(2.503e−02)	(1.725e+00)	(1.325e+00)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

Table 11
Mean best results of Ackley Function.

Dimension	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO-L	PSO.L+EES
2	1.501e+00	[a]**5.526e−02**	8.288e−05	**4.986e−09**	**7.994e−16**	1.155e−15	1.864e−01	**1.513e−07**
	(3.961e+00)	(2.305e−01)	(4.060e−04)	(2.611e−08)	(1.084e−15)	(1.445e−15)	(1.021e+00)	(2.356e−07)
5	**4.795e+00**	5.785e+00	1.455e+01	[a]**1.222e+01**	[a]**4.441e−16**	1.036e−15	1.182e+01	**9.777e+00**
	(5.543e+00)	(5.930e+00)	(3.460e+00)	(3.066e+00)	(0.000e+00)	(1.347e−15)	(7.040e+00)	(7.393e+00)
10	1.533e+01	**1.424e+01**	1.997e+01	**1.992e+01**	**4.441e−16**	5.625e−16	**1.909e+01**	2.003e+01
	(3.271e+00)	(3.891e+00)	(3.008e−01)	(2.796e−01)	(0.000e+00)	(6.486e−16)	(2.433e+00)	(1.515e+00)
20	**1.655e+01**	1.724e+01	2.026e+01	[a]**2.023e+01**	4.441e−16	4.441e−16	2.064e+01	**2.054e+01**
	(2.390e+00)	(1.833e+00)	(3.734e−02)	(3.825e−02)	(0.000e+00)	(0.000e+00)	(4.746e−01)	(8.279e−01)
30	1.752e+01	**1.745e+01**	2.045e+01	**2.043e+01**	4.441e−16	4.441e−16	[a]**2.094e+01**	2.101e+01
	(1.748e+00)	(2.014e+00)	(5.602e−02)	(5.939e−02)	(0.000e+00)	(0.000e+00)	(1.863e−01)	(6.879e−02)
40	**1.752e+01**	1.756e+01	**2.057e+01**	2.058e+01	4.441e−16	4.441e−16	**2.093e+01**	2.102e+01
	(1.896e+00)	(1.528e+00)	(5.234e−02)	(4.142e−02)	(0.000e+00)	(0.000e+00)	(3.013e−01)	(3.374e−01)
50	1.768e+01	**1.733e+01**	2.069e+01	[a]**2.067e+01**	4.441e−16	4.441e−16	[a]**2.103e+01**	2.113e+01
	(1.118e+00)	(1.709e+00)	(3.995e−02)	(3.057e−02)	(0.000e+00)	(0.000e+00)	(2.264e−01)	(1.169e−01)
60	[a]**1.737e+01**	1.795e+01	2.075e+01	**2.075e+01**	4.441e−16	4.441e−16	[a]**2.095e+01**	2.119e+01
	(1.377e+00)	(1.172e+00)	(3.594e−02)	(4.217e−02)	(0.000e+00)	(0.000e+00)	(4.278e−01)	(1.713e−01)
70	[a]**1.731e+01**	1.776e+01	**2.082e+01**	2.082e+01	4.441e−16	4.441e−16	[a]**2.094e+01**	2.120e+01
	(9.617e−01)	(9.475e−01)	(3.187e−02)	(4.743e−02)	(0.000e+00)	(0.000e+00)	(3.959e−01)	(2.238e−01)
80	**1.720e+01**	1.737e+01	2.087e+01	**2.086e+01**	4.441e−16	4.441e−16	[a]**2.094e+01**	2.118e+01
	(9.827e−01)	(9.810e−01)	(2.250e−02)	(2.433e−02)	(0.000e+00)	(0.000e+00)	(3.423e−01)	(2.129e−01)
90	**1.693e+01**	1.716e+01	**2.090e+01**	2.090e+01	4.441e−16	4.441e−16	[a]**2.098e+01**	2.120e+01
	(1.053e+00)	(1.049e+00)	(3.295e−02)	(2.735e−02)	(0.000e+00)	(0.000e+00)	(2.798e−01)	(2.614e−01)
100	**1.683e+01**	1.706e+01	2.095e+01	**2.094e+01**	4.441e−16	4.441e−16	[a]**2.093e+01**	2.119e+01
	(1.093e+00)	(7.936e−01)	(2.543e−02)	(3.035e−02)	(0.000e+00)	(0.000e+00)	(2.455e−01)	(3.095e−01)

Values in bold indicate the minimum result per algorithm in each dimension. Standard deviation in parentheses.
[a]Represents the minimum result based on the Wilcoxon signed rank test.

The results present in Tables 2, 3, 5, 6, 8, and 10 confirms that the EES provides an initial population that improves the performance of the evolutionary algorithms or, in the worst case, does not affect it.

4.2. Wilcoxon signed-rank test

In the experiments reported in Section 4, the same random population $\mathcal{P}_{0,r}$ was used as the initial population for every evolutionary algorithm (without and with EES) in each dimension for the run r. Namely, the rth run of each algorithm was initialized with the same values. The aim was to make a fair comparison against the evolutionary algorithms without EES and the evolutionary algorithms with EES.

In Tables 2–11, the $*$ is used to represent the version of each evolutionary algorithm (without and with EES) that is statistically better according to the Wilcoxon signed-rank test.

The Wilcoxon signed-rank test is a nonparametric statistical test that compares two paired groups. The test calculates the difference between sets of pairs and analyze these differences to establish whether they are statistically significant. In this paper, a significance level of 0.05 was compared against

Table 12
Wilcoxon signed-rank test for EMNA vs EMNA+EES.

Function	$EMNA < EMNA + EES$	$EMNA = EMNA + EES$	$EMNA > EMNA + EES$
StornChebyshev	0	1	1
InverseHilbert	0	9	0
LennardJones	2	9	1
Rastrigin	0	6	6
Griewangk	0	5	7
Weierstrass	0	11	1
ModifiedSchwefel	0	2	10
ExpandedSchaffer	0	12	0
HappyCat	0	5	7
Ackley	2	10	0
Total	4	70	33
%	3.74%	65.42%	30.84%

Table 13
Wilcoxon signed-rank test for DE vs DE+EES.

Function	$DE < DE + EES$	$DE = DE + EES$	$DE > DE + EES$
StornChebyshev	0	0	2
InverseHilbert	0	1	8
LennardJones	0	6	6
Rastrigin	2	1	9
Griewangk	0	3	9
Weierstrass	0	11	1
ModifiedSchwefel	0	1	11
ExpandedSchaffer	0	2	10
HappyCat	0	3	9
Ackley	0	7	5
Total	2	35	70
%	1.87%	32.71%	65.42%

the p-value obtained in each statistical test. A summary of the Wilcoxon signed-rank tests for each evolutionary algorithm, on each function, is shown in the Tables 12, 13, 14 and 15.

Table 12 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm. The Storn's Chebyshev Polynomial Fitting Problem and the Inverse Hilbert Matrix Problem were optimized only for a total of 2 and 9 different dimensions, respectively. The rest of the functions were optimized for a total of 12 dimensions. In total, 107 comparisons were development.

The second column of Table 12 summarizes the number of dimensions where the EES worsens the EMNA's performance. The third column of the table indicates the number of dimensions where the EES does not have an effect in the EMNA algorithm. The last column shows the number of dimensions where the EES improves the performance of EMNA. Only in 4 of the 107 (3.74%) tests, the EES worsens the performance of the EMNA algorithm. In 30.84% of the cases, the tests denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Modified Schwefel and Happy Cat functions, the EMNA with EES shows a relevant advantage over EMNA without EES. In the Inverse Hilbert, Lennard Jones, Weierstrass, Expanded Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA.

Table 13 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 65.42% of the tests, the explicit exploration strategy improves the DE's performance. Only in 2 dimensions of the Rastrigin function, the EES worsens the performance of the DE algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel, Expanded Schaffer and Happy Cat functions, the DE with EES shows a relevant advantage over DE without EES.

In Table 14 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 6 of the 107 tests (5.61%), the EES improves the performance of the PSO-G algorithm. In 25.23% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Lennard Jones and Weierstrass functions, the PSO-G with EES shows a significant disadvantage over PSO-G without EES.

In Table 15 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 53 of the 107 tests (49.53%), the EES improves the performance of the PSO-L algorithm. In 16.82% of the tests, the EES worsens the PSO-L's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions, the PSO-L with EES denotes an advantage over PSO-L without EES. For the Ackley function, the PSO-L with EES shows a significant disadvantage over PSO-L without EES.

4.3. Stop condition analysis

For denote the effect of the explicit exploration 5 different values $S = 5, 10, 15, 20, 25$ were tested like stop condition

Table 14
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES.

Function	$PSO-G < PSO-G+EES$	$PSO-G = PSO-G+EES$	$PSO-G > PSO-G+EES$
StornChebyshev	0	0	2
InverseHilbert	7	0	2
LennardJones	8	3	1
Rastrigin	0	12	0
Griewangk	0	12	0
Weierstrass	8	4	0
ModifiedSchwefel	0	12	0
ExpandedSchaffer	2	10	0
HappyCat	1	10	1
Ackley	1	11	0
Total	27	74	6
%	25.23%	69.16%	5.61%

Table 15
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES.

Function	$PSO-L < PSO.L+EES$	$PSO-L = PSO.L+EES$	$PSO-L > PSO.L+EES$
StornChebyshev	1	1	0
InverseHilbert	0	2	7
LennardJones	5	6	1
Rastrigin	1	1	10
Griewangk	0	2	10
Weierstrass	0	12	0
ModifiedSchwefel	1	0	11
ExpandedSchaffer	1	8	3
HappyCat	0	2	10
Ackley	9	2	1
Total	18	36	53
%	16.82%	33.64%	49.53%

Table 16
Wilcoxon signed-rank test for EMNA vs EMNA+EES with $S = 5$.

Function	$EMNA < EMNA+EES$	$EMNA = EMNA+EES$	$EMNA > EMNA+EES$
StornChebyshev	0	2	0
InverseHilbert	0	9	0
LennardJones	0	10	2
Rastrigin	0	8	4
Griewangk	1	5	6
Weierstrass	0	2	10
ModifiedSchwefel	0	2	10
ExpandedSchaffer	0	10	2
HappyCat	0	5	7
Ackley	1	6	5
Total	2	59	46
%	1.87%	55.14%	42.99%

criterion. If the best fitness value across S consecutive generations changes less than $1E-10$, the execution of the algorithm stops.

The best fitness value and generation were recorded throughout the 856 experiments with 30 runs executed for each stop condition value S. The comparison between different algorithm versions is based on the number of generations (including those from the EES), running the same number of fitness evaluations for evolutionary algorithms without EES and evolutionary algorithms with EES.

4.3.1. Stop condition $S = 5$

Table 16 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm with $S = 5$. Only in 2 of the 107 (1.87%) tests, the EES worsens the performance of the EMNA algorithm. In 42.99% of the cases, the tests denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Weierstrass, Modified Schwefel and Happy Cat functions, the EMNA with EES shows a relevant advantage over EMNA without EES. In the Inverse Hilbert, Lennard Jones, Rastrigin, Expanded Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA.

Table 17 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 93.46% of the tests, the explicit exploration strategy improves the DE's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones, Griewangk, Modified Schwefel and Expanded Schaffer functions, the DE with EES shows a relevant advantage over DE without EES.

In Table 18 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 11 of the 107 tests (10.28%), the EES improves the performance of the PSO-G algorithm. In 20.56% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones, Weierstrass and Expanded Schaffer functions, the PSO-G with EES shows a significant disadvantage over PSO-G without EES.

Table 17
Wilcoxon signed-rank test for DE vs DE+EES with $S = 5$.

Function	$DE < DE+EES$	$DE = DE+EES$	$DE > DE+EES$
StornChebyshev	0	0	2
InverseHilbert	0	1	8
LennardJones	0	0	12
Rastrigin	0	1	11
Griewangk	0	0	12
Weierstrass	0	2	10
ModifiedSchwefel	0	0	12
ExpandedSchaffer	0	0	12
HappyCat	0	1	11
Ackley	0	2	10
Total	0	7	100
%	0%	6.54%	93.46%

In Table 19 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 60 of the 107 tests (56.07%), the EES improves the performance of the PSO-L algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin and Weierstrass functions, the PSO-L with EES denotes an advantage over PSO-L without EES.

Table 20 shows the average number of executed generations until the best fitness value across $S = 5$ consecutive generations changes less than $1E-10$. In the entire benchmark, the 4 algorithms with EES run more generations than the algorithms without EES.

4.3.2. Stop condition $S = 10$

Table 21 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm with $S = 10$. Only in 2 of the 107 (1.87%) tests, the EES worsens the performance of the EMNA algorithm. In 37.38% of the cases, the tests

Table 18
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES with $S = 5$.

Function	$PSO-G < PSO-G+EES$	$PSO-G = PSO-G+EES$	$PSO-G > PSO-G+EES$
StornChebyshev	0	2	0
InverseHilbert	1	8	0
LennardJones	8	3	1
Rastrigin	0	10	2
Griewangk	0	12	0
Weierstrass	8	2	2
ModifiedSchwefel	0	10	2
ExpandedSchaffer	5	6	1
HappyCat	0	10	2
Ackley	0	11	1
Total	22	74	11
%	20.56%	69.16%	10.28%

Table 19
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES with $S = 5$.

Function	$PSO-L < PSO.L+EES$	$PSO-L = PSO.L+EES$	$PSO-L > PSO.L+EES$
StornChebyshev	0	0	2
InverseHilbert	0	0	9
LennardJones	0	6	6
Rastrigin	0	5	7
Griewangk	0	6	6
Weierstrass	0	1	11
ModifiedSchwefel	0	7	5
ExpandedSchaffer	0	8	4
HappyCat	0	6	6
Ackley	0	8	4
Total	0	47	60
%	0%	43.93%	56.07%

Table 20
Mean generations executed with stop condition $S = 5$.

Function	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO.L	PSO.L+EES
StornChebyshev	56.27	77.32	14.95	35.15	25.8	49.07	34.57	59.22
InverseHilbert	77.55	88.7	17.87	29.73	44.2	51.49	37.88	48.32
LennardJones	13.25	40.37	14.07	42.61	11.88	40.54	11.97	40.46
Rastrigin	48.45	59.31	19.05	37.02	38.92	53.36	51.37	63.39
Griewangk	101.74	112.97	17.77	36.17	32.52	44.39	51.62	63.67
Weierstrass	10.91	19.38	21.89	30.57	15.25	21.97	10.81	19.57
ModifiedSchwefel	116.79	123.19	24.89	41.9	34.87	48.26	56.54	68.56
ExpandedSchaffer	12.38	22.05	20.28	28.77	33.43	40.93	11.96	21.25
HappyCat	108.03	114.19	19.43	36.9	22.56	35.37	58.05	68.96
Ackley	15.78	22.24	20.35	28.99	44.17	57.98	12.07	19.78

Table 21
Wilcoxon signed-rank test for EMNA vs EMNA+EES with $S = 10$.

Function	$EMNA < EMNA+EES$	$EMNA = EMNA+EES$	$EMNA > EMNA+EES$
StornChebyshev	0	1	1
InverseHilbert	0	9	0
LennardJones	1	9	2
Rastrigin	0	8	4
Griewangk	0	5	7
Weierstrass	0	7	5
ModifiedSchwefel	0	2	10
ExpandedSchaffer	0	10	2
HappyCat	0	5	7
Ackley	1	9	2
Total	2	65	40
%	1.87%	60.75%	37.38%

denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Modified Schwefel and Happy Cat functions, the EMNA+EES algorithm shows a relevant advantage over EMNA algorithm. In the Inverse Hilbert, Lennard Jones, Rastrigin, Expanded Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA algorithm.

Table 22 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 58.88% of the tests, the explicit exploration strategy improves the DE's performance. In only 1 dimension for the Rastrigin function, the EES worsens the average results of the DE algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Lennard Jones, Rastrigin, Modified Schwefel and Expanded Schaffer functions, the DE with EES shows a relevant advantage over DE without EES.

Table 22
Wilcoxon signed-rank test for DE vs DE+EES with $S = 10$.

Function	$DE < DE + EES$	$DE = DE + EES$	$DE > DE + EES$
StornChebyshev	0	0	2
InverseHilbert	0	3	6
LennardJones	0	3	9
Rastrigin	1	4	7
Griewangk	0	7	5
Weierstrass	0	10	2
ModifiedSchwefel	0	1	11
ExpandedSchaffer	0	1	11
HappyCat	0	6	6
Ackley	0	8	4
Total	1	43	63
%	0.93%	40.19%	58.88%

In Table 23 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 8 of the 107 tests (7.48%), the EES improves the performance of the PSO-G algorithm. In 22.43% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones and Weierstrass functions, the PSO-G+EES algorithm shows a significant disadvantage over PSO-G without EES.

In Table 24 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 53 of the 107 tests (49.53%), the EES improves the performance of the PSO-L algorithm. In only 4 dimensions for the Ackley function, the EES worsens the average results of the DE algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk Modified Schwefel and Happy Cat functions, the PSO-L+EES algorithm denotes an advantage over PSO-L algorithm.

Table 25 shows the average number of executed generations until the best fitness value across $S = 10$ consecutive generations changes less than $1E - 10$. In the entire benchmark, the 4 algorithms with EES run more generations than the 4 algorithms without EES.

4.3.3. Stop condition $S = 15$

Table 26 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm with $S = 15$. Only in 5 of the 107 (4.67%) tests, the EES worsens the performance of the EMNA algorithm. In 32.71% of the cases, the tests denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Modified Schwefel and Happy Cat functions, the EMNA+EES algorithm shows a relevant advantage over EMNA algorithm. In the Inverse Hilbert, Rastrigin, Expanded Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA algorithm.

Table 27 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 61.68% of the tests, the explicit exploration strategy improves the DE's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Lennard Jones, Rastrigin, Griewangk, Modified Schwefel, Expanded Schaffer and Happy Cat functions, the DE with EES shows a relevant advantage over DE without EES. In the Weierstrass and Ackley functions, the EES neither represents an advantage nor a disadvantage for the DE algorithm.

In Table 28 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 4 of the 107 tests (3.74%), the EES improves the performance of the PSO-G algorithm. In 17.76% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones and Weierstrass functions, the PSO-G+EES algorithm shows a significant disadvantage over PSO-G without EES. For all other functions, the EES is neither an advantage nor a disadvantage for the DE algorithm.

Table 23
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES with $S = 10$.

Function	$PSO - G < PSO - G + EES$	$PSO - G = PSO - G + EES$	$PSO - G > PSO - G + EES$
StornChebyshev	0	0	2
InverseHilbert	1	8	0
LennardJones	8	3	1
Rastrigin	1	11	0
Griewangk	0	10	2
Weierstrass	11	0	1
ModifiedSchwefel	2	9	1
ExpandedSchaffer	0	12	0
HappyCat	1	11	0
Ackley	0	11	1
Total	24	75	8
%	22.43%	70.09%	7.48%

Table 24
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES with $S = 10$.

Function	$PSO - L < PSO.L + EES$	$PSO - L = PSO.L + EES$	$PSO - L > PSO.L + EES$
StornChebyshev	0	1	1
InverseHilbert	0	3	6
LennardJones	0	10	2
Rastrigin	0	3	9
Griewangk	0	4	8
Weierstrass	0	7	5
ModifiedSchwefel	0	3	9
ExpandedSchaffer	0	9	3
HappyCat	0	3	9
Ackley	4	7	1
Total	4	50	53
%	3.74%	46.73%	49.53%

Table 25
Mean generations executed with stop condition $S = 10$.

Function	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO.L	PSO.L+EES
StornChebyshev	110.18	129.58	29.77	50.75	43.57	66.38	120.18	133.63
InverseHilbert	89.64	98.95	39.96	47.18	100.83	109.83	122.94	130.05
LennardJones	25.42	51.6	34.26	61.4	21.07	50.02	24.7	50.48
Rastrigin	75.97	88.01	81.19	102.7	64.11	76.78	115.41	121.81
Griewangk	117.03	125.92	76.04	93.26	48.72	60.84	117.48	120.36
Weierstrass	20.08	26.44	55.4	60.15	69.3	60.64	20.01	26.44
ModifiedSchwefel	125.09	131.69	72.69	94.92	50.98	62.9	132.12	137.22
ExpandedSchaffer	24.04	33.09	49.87	58.65	59.48	70.98	20.14	29.41
HappyCat	119.21	124.53	80.41	97.26	42.13	54.89	129.78	132.74
Ackley	31.52	37.61	54	60.37	88.64	98.83	28.81	33.54

Table 26
Wilcoxon signed-rank test for EMNA vs EMNA+EES with $S = 15$.

Function	$EMNA < EMNA + EES$	$EMNA = EMNA + EES$	$EMNA > EMNA + EES$
StornChebyshev	0	1	1
InverseHilbert	0	9	0
LennardJones	4	6	2
Rastrigin	0	11	1
Griewangk	0	5	7
Weierstrass	0	7	5
ModifiedSchwefel	0	2	10
ExpandedSchaffer	0	10	2
HappyCat	0	5	7
Ackley	1	11	0
Total	5	67	35
%	4.67%	62.62%	32.71%

Table 27
Wilcoxon signed-rank test for DE vs DE+EES with $S = 15$.

Function	$DE < DE + EES$	$DE = DE + EES$	$DE > DE + EES$
StornChebyshev	0	0	2
InverseHilbert	0	3	6
LennardJones	0	2	10
Rastrigin	0	4	8
Griewangk	0	3	9
Weierstrass	0	11	1
ModifiedSchwefel	0	1	11
ExpandedSchaffer	0	3	9
HappyCat	0	4	8
Ackley	0	10	2
Total	0	41	66
%	0%	38.32%	61.68%

In Table 29 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 55 of the 107 tests (51.4%), the EES improves the performance of the PSO-L algorithm. In only 10 of the 107 tests (9.35%), the EES worsens the average results of the DE algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions, the PSO-L+EES algorithm denotes an advantage over PSO-L algorithm. In the Ackley function, the PSO-L+EES algorithm shows a significant disadvantage over PSO-L without EES.

Table 30 shows the average number of executed generations until the best fitness value across $S = 15$ consecutive generations changes less than $1E - 10$. The EMNA, DE and PSO-G algorithms run less generations than the EMNA+EES, DE+EES and PSO-G+EES algorithms, respectively. In the case of PSO-L versus PSO-L+EES, the difference is insignificant in several functions.

4.3.4. Stop condition $S = 20$

Table 31 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm with $S = 20$. Only in 2 of the 107 (1.87%) tests, the EES worsens the performance of the EMNA algorithm. In 30.84% of the cases, the tests denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Modified Schwefel and Happy Cat functions, the EMNA+EES algorithm shows a relevant advantage over EMNA algorithm. In the Inverse Hilbert, Lennard Jones, Rastrigin, Weierstrass, Expanded Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA algorithm.

Table 32 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 65.42% of the tests, the explicit exploration strategy improves the DE's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Lennard Jones, Rastrigin, Griewangk, Modified Schwefel, Expanded Schaffer and Happy Cat functions, the DE with EES shows a relevant advantage over DE without EES. In the Weierstrass and Ackley functions, the EES neither represents an advantage nor a disadvantage for the DE algorithm.

In Table 33 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 7 of the 107 tests (6.54%), the EES improves the performance of the PSO-G algorithm. In 19.63% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones and Weierstrass functions, the PSO-G+EES algorithm shows a significant disadvantage over PSO-G without EES. For all other functions, the EES is neither an advantage nor a disadvantage for the PSO-G algorithm.

In Table 34 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 53 of the 107 tests (49.53%), the EES improves the performance of the PSO-L algorithm. In only 12 of the 107 tests (11.21%), the EES worsens the average results of the PSO-L algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions, the PSO-L+EES algorithm denotes an advantage over PSO-L algorithm. In the Ackley function, the PSO-L+EES algorithm shows a significant disadvantage over PSO-L without EES.

Table 28
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES with $S = 15$.

Function	$PSO-G < PSO-G+EES$	$PSO-G = PSO-G+EES$	$PSO-G > PSO-G+EES$
StornChebyshev	0	0	2
InverseHilbert	0	9	0
LennardJones	8	3	1
Rastrigin	1	10	1
Griewangk	0	12	0
Weierstrass	8	4	0
ModifiedSchwefel	1	11	0
ExpandedSchaffer	1	11	0
HappyCat	0	12	0
Ackley	0	12	0
Total	19	84	4
%	17.76%	78.5%	3.74%

Table 29
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES with $S = 15$.

Function	$PSO-L < PSO.L+EES$	$PSO-L = PSO.L+EES$	$PSO-L > PSO.L+EES$
StornChebyshev	1	1	0
InverseHilbert	1	1	7
LennardJones	1	8	3
Rastrigin	0	2	10
Griewangk	0	2	10
Weierstrass	0	11	1
ModifiedSchwefel	0	2	10
ExpandedSchaffer	1	9	2
HappyCat	0	2	10
Ackley	6	4	2
Total	10	42	55
%	9.35%	39.25%	51.4%

Table 30
Mean generations executed with stop condition $S = 15$.

Function	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO.L	PSO.L+EES
StornChebyshev	131	147.38	41.92	65.15	57	81.88	145.97	145.72
InverseHilbert	97.81	107.04	66.37	71.4	111.59	121.63	145.88	146.14
LennardJones	37.59	63.24	58.02	85.96	30.72	58.62	37.42	60.29
Rastrigin	111.47	118.15	132.86	139.12	70.3	82.92	132.32	135.85
Griewangk	124.96	131.16	118.75	129.06	54.94	67.1	126.2	130.66
Weierstrass	29.43	34.26	92.05	94.89	112.57	104.11	30.06	33.83
ModifiedSchwefel	128.46	133.92	113.04	127.98	56.56	68.46	143.04	145.96
ExpandedSchaffer	37.27	46.88	90.09	93.53	68.71	79.79	28.72	36.69
HappyCat	128.27	132.73	124.78	132.76	58.23	70.8	140.21	144.06
Ackley	48.34	51.35	91.21	93.25	96.87	107.2	45.04	47.61

Table 31
Wilcoxon signed-rank test for EMNA vs EMNA+EES with $S = 20$.

Function	$EMNA < EMNA+EES$	$EMNA = EMNA+EES$	$EMNA > EMNA+EES$
StornChebyshev	0	1	1
InverseHilbert	0	9	0
LennardJones	0	10	2
Rastrigin	1	8	3
Griewangk	0	5	7
Weierstrass	0	10	2
ModifiedSchwefel	0	2	10
ExpandedSchaffer	0	11	1
HappyCat	0	5	7
Ackley	1	11	0
Total	2	72	33
%	1.87%	67.29%	30.84%

Table 35 shows the average number of executed generations until the best fitness value across $S = 20$ consecutive generations changes less than $1E-10$. In general, the 4 algorithms with EES execute more generations than the 4 algorithms without EES. However, the difference is insignificant in several functions.

4.3.5. Stop condition $S = 25$

Table 36 presents the Wilcoxon test results of the 10 functions for the Estimation of Multivariate Normal Algorithm with $S = 25$. Only in 4 of the 107 (3.74%) tests, the EES worsens the performance of the EMNA algorithm. In 34.58% of the cases, the tests denote a significant improvement of the EMNA's performance applying the explicit exploration strategy. In the remaining tests, there is not a statistical difference. Particularly, in the Griewangk, Modified Schwefel and Happy Cat functions, the EMNA+EES algorithm shows a relevant advantage over EMNA algorithm. In the Inverse Hilbert, Lennard Jones, Rastrigin, Weierstrass, Expanded

Table 32
Wilcoxon signed-rank test for DE vs DE+EES with $S = 20$.

Function	$DE < DE + EES$	$DE = DE + EES$	$DE > DE + EES$
StornChebyshev	0	0	2
InverseHilbert	0	2	7
LennardJones	0	3	9
Rastrigin	0	3	9
Griewangk	0	3	9
Weierstrass	0	9	3
ModifiedSchwefel	0	1	11
ExpandedSchaffer	0	3	9
HappyCat	0	3	9
Ackley	0	10	2
Total	0	37	70
%	0%	34.58%	65.42%

Schaffer and Ackley functions, the EES neither represents an advantage nor a disadvantage for the EMNA algorithm.

Table 37 presents the Wilcoxon test results of the 10 functions for the Differential Evolution algorithm. In 63.55% of the tests, the explicit exploration strategy improves the DE's performance. In only 1 dimension for the Rastrigin function, the EES worsens the average results of the DE algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel, Expanded Schaffer and Happy Cat functions, the DE+EES shows a relevant advantage over DE without EES. In the Weierstrass and Ackley functions, the EES neither represents an advantage nor a disadvantage for the DE algorithm.

In Table 38 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Star Neighborhood algorithm are shown. Only in 6 of the 107 tests (5.61%), the EES improves the performance of the PSO-G algorithm. In 20.56% of the tests, the EES worsens the PSO-G's performance. In the remaining tests, there is not a statistical difference. Particularly, in the Lennard Jones and Weierstrass functions, the PSO-G+EES algorithm shows a significant disadvantage over PSO-G without EES. For all other functions, the EES is neither an advantage nor a disadvantage for the PSO-G algorithm.

In Table 39 the Wilcoxon test results of the 10 functions for the Particle Swarm Optimization with Ring Neighborhood algorithm are shown. In 54 of the 107 tests (50.47%), the EES improves the performance of the PSO-L algorithm. In only 10 of the 107 tests (9.35%), the EES worsens the average results of the PSO-L algorithm. In the remaining tests, there is not a statistical difference. Particularly, in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions, the PSO-L+EES algorithm denotes an advantage over PSO-L algorithm. In the Ackley function, the PSO-L+EES algorithm shows a significant disadvantage over PSO-L without EES.

Table 40 shows the average number of executed generations until the best fitness value across $S = 25$ consecutive generations changes less than $1E - 10$. In general, the 4 algorithms with EES execute more generations than the 4 algorithms without EES. However, the difference is insignificant in several functions. Note that in the case of PSO-L and PSO-L+EES the maximum number of generations was reached for the Storn Chebyshev and Inverse Hilbert.

5. Conclusions

The evidence provided by the experimental results shows that EES can significantly improve the effectiveness of Evolutionary Algorithms or at least it is not a disadvantage. In general, the effectiveness of the PSO-L, EMNA and DE algorithms is improved with the incorporation of the EES. On the other hand, the effectiveness of the PSO-G remains unchanged with the inclusion of the EES; in most tests.

Besides the search paradigm of the evolutionary algorithm, the performance achieved by the incorporation of the EES is also problem dependent. The results obtained from the Rastrigin, Griewangk, Modified Schwefel, Expanded Schaffer and Happy Cat functions are improved when the EES is included in the EMNA algorithm. In the case of DE algorithm, the results are improved in the Inverse Hilbert, Lennard Jones, Rastrigin, Griewangk, Modified

Table 33
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES with $S = 20$.

Function	$PSO - G < PSO - G + EES$	$PSO - G = PSO - G + EES$	$PSO - G > PSO - G + EES$
StornChebyshev	0	0	2
InverseHilbert	0	7	2
LennardJones	8	3	1
Rastrigin	0	11	1
Griewangk	2	10	0
Weierstrass	9	3	0
ModifiedSchwefel	0	12	0
ExpandedSchaffer	1	11	0
HappyCat	0	12	0
Ackley	1	10	1
Total	21	79	7
%	19.63%	73.83%	6.54%

Table 34
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES with $S = 20$.

Function	$PSO - L < PSO.L + EES$	$PSO - L = PSO.L + EES$	$PSO - L > PSO.L + EES$
StornChebyshev	1	1	0
InverseHilbert	0	2	7
LennardJones	3	8	1
Rastrigin	0	3	9
Griewangk	0	2	10
Weierstrass	0	12	0
ModifiedSchwefel	0	1	11
ExpandedSchaffer	2	7	3
HappyCat	0	1	11
Ackley	6	5	1
Total	12	42	53
%	11.21%	39.25%	49.53%

Table 35
Mean generations executed with stop condition $S = 20$.

Function	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO.L	PSO.L+EES
StornChebyshev	138.65	150	59.97	80.65	68.55	94.93	150	150
InverseHilbert	103.95	112.93	88.33	97.06	117.33	128.87	150	149.58
LennardJones	46.24	72.2	84.08	106.75	39.82	67.15	49.8	70.47
Rastrigin	128.01	132.38	144.68	146.59	75.3	87.98	140.56	141.13
Griewangk	129.17	135.6	134.24	139.62	60.49	73.21	131.33	134.98
Weierstrass	39.37	42.42	114.67	118.53	130.05	128.62	39.4	41.74
ModifiedSchwefel	130.79	135.76	132.09	141.26	61.62	73.46	145.68	147.56
ExpandedSchaffer	50.89	60.18	116.43	118.12	74.58	85.54	36.6	44.36
HappyCat	135.9	139.57	135.07	140.14	71.26	86.17	144.29	147.51
Ackley	67.21	67.97	116.36	119.9	102.17	112.2	62.13	59.49

Table 36
Wilcoxon signed-rank test for EMNA vs EMNA+EES with $S = 25$.

Function	$EMNA < EMNA + EES$	$EMNA = EMNA + EES$	$EMNA > EMNA + EES$
StornChebyshev	0	1	1
InverseHilbert	0	9	0
LennardJones	0	10	2
Rastrigin	0	8	4
Griewangk	0	5	7
Weierstrass	1	8	3
ModifiedSchwefel	0	2	10
ExpandedSchaffer	1	9	2
HappyCat	0	4	8
Ackley	2	10	0
Total	4	66	37
%	3.74%	61.68%	34.58%

Table 37
Wilcoxon signed-rank test for DE vs DE+EES with $S = 25$.

Function	$DE < DE + EES$	$DE = DE + EES$	$DE > DE + EES$
StornChebyshev	0	0	2
InverseHilbert	0	0	9
LennardJones	0	6	6
Rastrigin	1	2	9
Griewangk	0	3	9
Weierstrass	0	12	0
ModifiedSchwefel	0	1	11
ExpandedSchaffer	0	2	10
HappyCat	0	3	9
Ackley	0	9	3
Total	1	38	68
%	0.93%	35.51%	63.55%

Schwefel, Expanded Schaffer and Happy Cat functions when the EES is included. The PSO-G algorithm performs better without EES in the Inverse Hilbert, Lennard Jones and Weierstrass functions. In the same way, the PSO-L algorithm works better without EES in the Lennard Jones and Ackley functions; although the PSO-L+EES algorithm works better in the Inverse Hilbert, Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions.

In general, the EES provides an advantage for the Rastrigin, Griewangk, Modified Schwefel and Happy Cat functions. Nevertheless, in the case of the Lennard Jones and Weierstrass functions the Explicit Exploration Strategy affects the performance of the algorithms. For the Storn Chebyshev, Inverse Hilbert, Expanded Schaffer and Ackley functions, the results are not conclusive for the EES.

On the other hand, when we add the stop condition, the EES denotes better performance when $S = 5$. In this case, the DE+EES algorithm outperforms the DE algorithm in the whole benchmark. The PSO-L+EES algorithm shows a statistical difference in 7 of 10 functions. The EMNA+EES algorithm obtains the best results in 4 out of 10 functions. Only the PSO-G+EES algorithm presents a significant disadvantage in 3 of the 10 functions. For a stop condition $S \geq 10$, the effect of EES on the algorithms decreases, but its advantage remains for the EMNA, DE, and PSO-L algorithms. The EES does not imply to modify the main search paradigm of the evolutionary algorithms and can be easily implemented. Thus, it has the potential of being used jointly in a wide range of evolutionary algorithms.

One contribution of this paper is the directed attention to the initial exploration. An important source of population diversity is the initial population. The EES proposes to decrease the risk of

Table 38
Wilcoxon signed-rank test for PSO-G vs PSO-G+EES with $S = 25$.

Function	$PSO-G < PSO-G+EES$	$PSO-G = PSO-G+EES$	$PSO-G > PSO-G+EES$
StornChebyshev	0	0	2
InverseHilbert	1	6	2
LennardJones	8	3	1
Rastrigin	0	12	0
Griewangk	0	12	0
Weierstrass	8	4	0
ModifiedSchwefel	1	10	1
ExpandedSchaffer	2	10	0
HappyCat	0	12	0
Ackley	2	10	0
Total	22	79	6
%	20.56%	73.83%	5.61%

Table 39
Wilcoxon signed-rank test for PSO-L vs PSO.L+EES with $S = 25$.

Function	$PSO-L < PSO.L+EES$	$PSO-L = PSO.L+EES$	$PSO-L > PSO.L+EES$
StornChebyshev	1	1	0
InverseHilbert	0	2	7
LennardJones	2	9	1
Rastrigin	0	2	10
Griewangk	0	2	10
Weierstrass	0	12	0
ModifiedSchwefel	0	1	11
ExpandedSchaffer	1	8	3
HappyCat	0	2	10
Ackley	6	4	2
Total	10	43	54
%	9.35%	40.19%	50.47%

Table 40
Mean generations executed with stop condition $S = 25$.

Function	EMNA	EMNA+EES	DE	DE+EES	PSO-G	PSO-G+EES	PSO.L	PSO.L+EES
StornChebyshev	144.77	150	75.65	97.4	80.1	106.18	150	150
InverseHilbert	109.51	118.69	108.16	114.61	123.23	135.03	150	150
LennardJones	55.22	81.21	107.78	120.15	47.43	75.96	60.44	79.7
Rastrigin	134.38	139.74	146.34	148.58	80.3	92.98	143.84	144.02
Griewangk	132.75	138.04	140.55	142.87	65.69	78.54	135	137.85
Weierstrass	48.88	51.78	131.02	130.69	137.26	138.6	48.37	49.49
ModifiedSchwefel	132.59	137.42	140.22	145.7	66.62	78.46	146.41	148.31
ExpandedSchaffer	63.89	75.67	130.71	132.33	79.93	91.56	43.45	51.67
HappyCat	140.18	143.88	138.64	143.19	85.31	100.19	147.27	148.66
Ackley	88.44	82.65	129.7	132.65	107.17	117.2	74.69	68.26

premature convergence and to improve the effectiveness of the EAs by means of the initial exploration.

The EES uses as stopping criterion the information provided by the fitness distribution. In general, for a high dimensional search space, a few number of generations for explicit exploration are enough to provide adequate information about the fitness distribution because the fitness refers to a single variable.

The EES can be considered as a tactical strategy specialized in unbiased and random search. This approach may receive more attention in future works and can be used for providing a clear diagnosis about the difficulty level of the optimization problems.

CRediT authorship contribution statement

Rogelio Salinas-Gutiérrez: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing, Visualization, Project administration. **Angel Eduardo Muñoz Zavala:** Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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